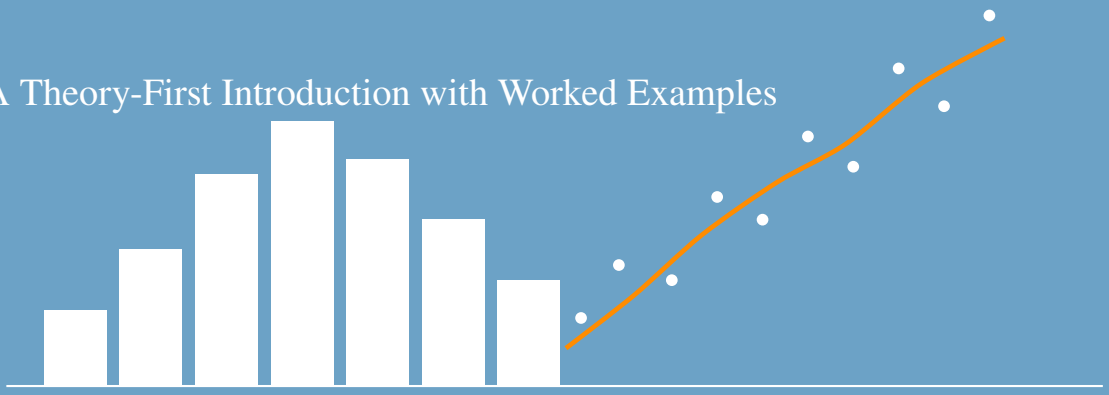
Statistics and Data Analysis for Engineers

A Theory-First Introduction with Worked Examples



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Part I

Data and Probability Models

Chapter 1 Descriptive Statistics

In software engineering, we constantly deal with data: performance metrics, user behavior patterns, system response times, and code complexity measures. To make sense of this information and extract meaningful insights, we need tools to summarize, organize, and visualize data. Descriptive statistics provides these essential tools, allowing us to understand the characteristics of our data without making inferences about larger populations.

Consider a software development team analyzing the performance of their web application. They collect response times for 1000 user requests and find values ranging from 50 milliseconds to 3.2 seconds. Without proper analysis, this raw data is overwhelming and uninformative. However, by applying descriptive statistics, they can determine that the average response time is 450 milliseconds, that 75% of requests complete within 600 milliseconds, and that there are a few unusually slow requests that might indicate performance issues.

This chapter introduces the fundamental concepts of descriptive statistics, focusing on measures that help us understand the central tendencies, variability, and distribution patterns in our data. We will explore how these techniques apply specifically to software engineering contexts, from analyzing algorithm performance to understanding user behavior patterns.

1.1 Introduction to Descriptive Statistics

Sometimes statistical work begins with existing data, such as precipitation records, unemployment rates, or GDP figures, which we then summarize and analyze. In other situations, data must be generated through an experiment or study. For example, to compare two teaching methods in an introductory programming course, an instructor might randomly divide students into two groups, apply a different method to each, and then compare test scores. Random assignment is essential: it ensures that differences between groups are not due to preexisting factors such as aptitude, but rather to the teaching method itself.

After the data are collected, they are summarized and visualized—for example, by reporting the average score for each group. This process is the essence of **descriptive statistics**: organising, presenting, and describing data in a meaningful way.

Population versus Sample

In statistical analysis, we distinguish between two key concepts:

Definition 1.1 (Population)

A **population** is the complete set of all possible observations or measurements of interest in a particular study. In software engineering contexts, this might include all possible execution times of an algorithm, all user sessions on a website, or all lines of code in a project. ♣

Definition 1.2 (Sample)

A **sample** is a subset of the population that we actually observe or measure. Due to practical constraints, we often work with samples rather than entire populations. ♣

In the teaching-method example, the students in the classroom form only a *sample* of the larger *population* of potential learners. Descriptive statistics helps us understand the sample itself, while inferential methods later allow us to use that sample to draw conclusions about the broader population. Inferential statistics lies beyond the scope of this chapter and is developed later in this book. We focus here on descriptive statistics.

Example 1.1 Software Performance Analysis

Consider a web application serving millions of users daily. The **population** would be the response times for all possible user requests. However, due to computational and storage limitations, we might only collect response times for a **sample** of 10,000 requests per day. This sample should be representative of the population to draw meaningful conclusions.

Types of Data

Data can be classified into different types, each requiring different statistical approaches:

Definition 1.3 (Qualitative Data)

Qualitative data (also called categorical data) consists of non-numerical information that can be categorized. Examples include programming languages used in a project, user satisfaction ratings (satisfied/neutral/dissatisfied), or bug severity levels (critical/high/medium/low).

♣

The essential difference between **qualitative** and **quantitative** data lies in what they represent and how they are analyzed. Qualitative data refers to categories or attributes that cannot be expressed meaningfully as numbers. Such data are used to classify or label elements, for example, programming language or bug severity. Quantitative data, on the other hand, consists of numerical values that represent counts or measurements, such as response times or number of users. Because these values are numerical, they permit mathematical operations and statistical analysis. Recognizing the distinction between qualitative and quantitative data is fundamental, as it guides the choice of statistical methods and the types of visualizations that are appropriate for analysis.

Definition 1.4 (Quantitative Data)

Quantitative data consists of numerical measurements that can be ordered and subjected to mathematical operations. This can be further divided into:

- **Discrete**: Countable values (number of bugs, lines of code, user sessions)
- **Continuous**: Measurable values that can take any value within a range (response times, memory usage, CPU utilization)

♣

Example 1.2

- **Qualitative**: Programming language (Java, C++, Rust), deployment environment (development, staging, production)
- **Quantitative Discrete**: Number of commits per day, lines of code, number of test cases
- **Quantitative Continuous**: Response time in milliseconds, memory usage in MB, CPU utilization percentage

Outliers and Extreme Data

In any dataset, most values tend to cluster around a central region, but occasionally, we encounter values that are much higher or lower than the rest. These are called **outliers** or **extreme values**. Outliers can arise for many reasons: measurement errors, unusual but valid events, or natural variability in the data.

For example, if most web requests complete in under 400 milliseconds, but one request takes 1200 milliseconds, that 1200ms value is an outlier. Outliers can have a strong influence on summary statistics like the mean, making the data appear more variable or shifting the average away from where most values lie.

It's important to look for outliers when analyzing data, as they can signal interesting phenomena (such as a rare performance bottleneck), data entry mistakes, or the need for further investigation. While there are formal methods to detect outliers, which we will discuss later, often a simple plot or a scan of the sorted data is enough to spot values that "stand out" from the rest.

In practice, understanding the context is key: sometimes outliers are errors to be corrected or removed, but other times they are the most important part of the story.

1.2 Describing Data Sets

When presenting numerical results, clarity and brevity are essential. Tables and graphs are particularly effective, as they allow the reader to quickly grasp the main characteristics of a dataset. Visual and tabular summaries often highlight important aspects such as the overall range, how tightly the values are concentrated, and whether the data appear symmetric or skewed.

Frequency Tables and Graphs

The first step in data analysis is often to organize raw data into a frequency distribution, which shows how often each value (or range of values) occurs.

Definition 1.5 (Frequency Distribution)

A **frequency distribution** is a table that shows the frequency (count) of each value or class of values in a dataset. It can be presented as:

- **Absolute frequency**: The actual count of occurrences
- **Relative frequency**: The proportion of total observations
- **Cumulative frequency**: The running total of frequencies



Example 1.3 A dataset with a relatively small number of distinct values can be conveniently presented in a frequency table. For instance, consider a dataset consisting of the starting monthly salaries (to the nearest thousand Danish Kroner) of 42 recently graduated students with B.S. degrees in software engineering.

The raw data is as follows:

$$37, 37, 37, 37, 38, 39, 39, 39, 40, 40, 40, 40, 40, 41, 41, 41, 41, 41, 41, 41, 41,$$
$$42, 42, 42, 42, 42, 42, 42, 42, 42, 42, 44, 44, 44, 44, 44, 46, 46, 47, 47, 47, 50$$

From this data, we can construct a frequency table. Table 1.1 tells us, among other things, that the lowest starting salary of 37,000 DKK was received by four of the graduates, whereas the highest salary of 50,000 DKK

was received by a single student. The most common starting salary was 42,000 DKK, received by 10 of the students.

Salary ($\times$1000 **DKK**)	**Frequency**
37	4
38	1
39	3
40	5
41	8
42	10
44	5
46	2
47	3
50	1

Table 1.1: Frequency Distribution of Monthly Starting Salaries

Data from a frequency table can be graphically represented by a line graph that plots the distinct data values on the horizontal axis and indicates their frequencies by the heights of vertical lines. A line graph of the data presented in Table 1.1 is shown in Figure 1.1.

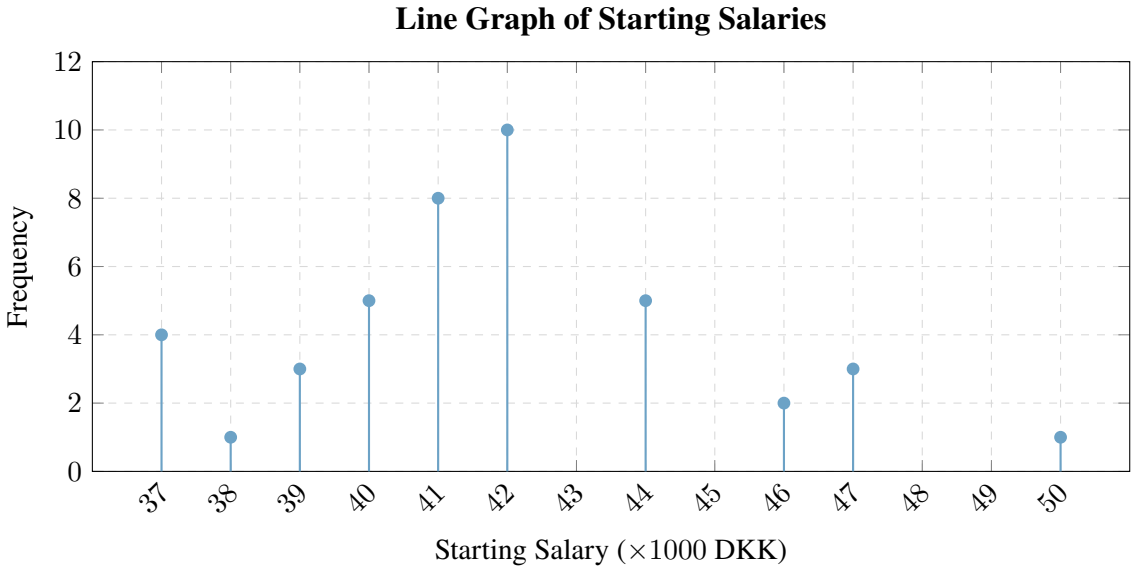


Figure 1.1: A line graph showing the frequency of different starting salaries.

When the lines in a line graph are given added thickness, the graph is called a bar graph. Figure 1.2 shows a bar graph for the same salary data.

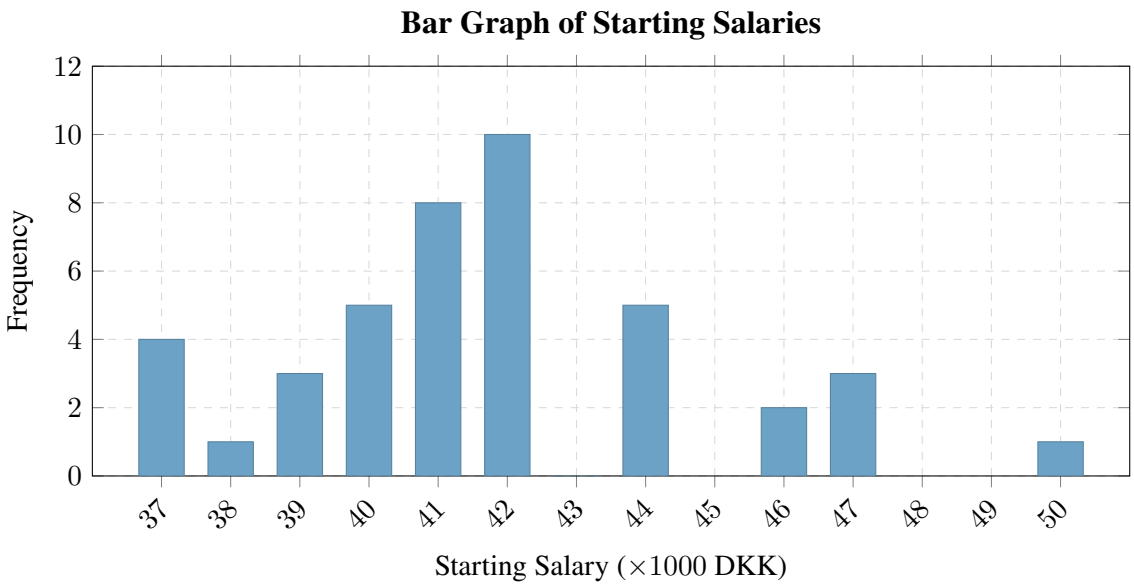


Figure 1.2: A bar graph of the salary data.

Another type of graph used to represent a frequency table is the frequency polygon, which plots the frequencies of the different data values on the vertical axis, and then connects the plotted points with straight lines. Figure 1.3 shows a frequency polygon for the same salary data.

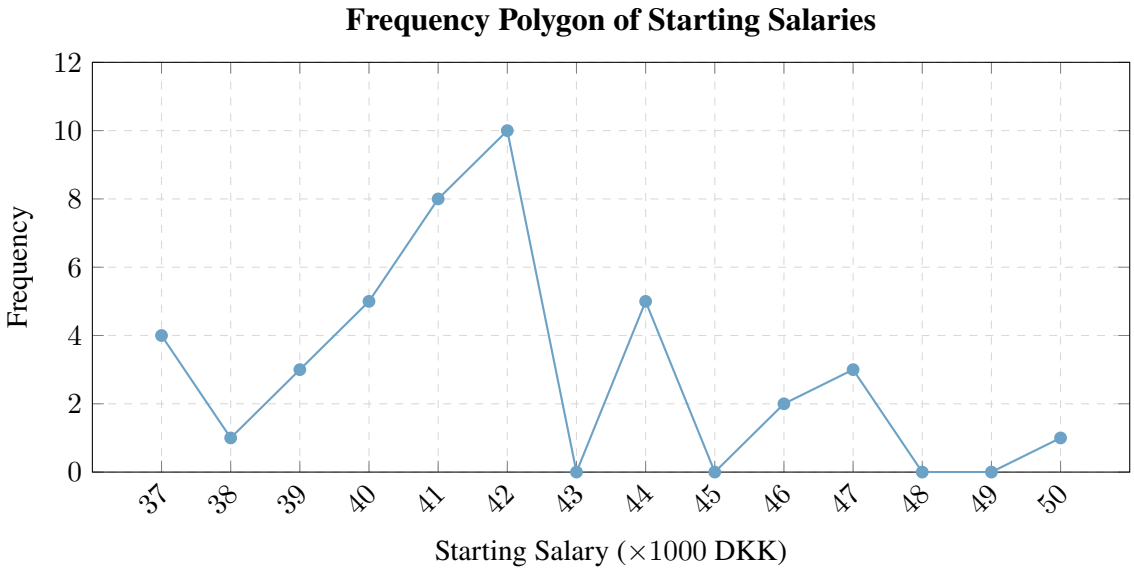


Figure 1.3: A frequency polygon of the salary data, connecting the points from the frequency table.

Sometimes we prefer the relative frequency over the absolute frequency. This is particularly useful when dealing with large datasets and when you want to compare values from datasets of different sizes. An extension of the relative frequency is the cumulative relative frequency, which is the running total of the relative frequencies. Table 1.2 shows the frequency, relative frequency, and cumulative relative frequency for the salary data.

Salary	Frequency	Relative Freq.	Cumulative Rel. Freq.
37	4	$4/42 \approx 0.095$	$4/42 \approx 0.095$
38	1	$1/42 \approx 0.024$	$5/42 \approx 0.119$
39	3	$3/42 \approx 0.071$	$8/42 \approx 0.190$
40	5	$5/42 \approx 0.119$	$13/42 \approx 0.310$
41	8	$8/42 \approx 0.190$	$21/42 = 0.500$
42	10	$10/42 \approx 0.238$	$31/42 \approx 0.738$
44	5	$5/42 \approx 0.119$	$36/42 \approx 0.857$
46	2	$2/42 \approx 0.048$	$38/42 \approx 0.905$
47	3	$3/42 \approx 0.071$	$41/42 \approx 0.976$
50	1	$1/42 \approx 0.024$	$42/42 = 1.000$

Table 1.2: Frequency, Relative Frequency, and Cumulative Relative Frequency of Monthly Starting Salaries.

Grouped Data and Histograms

As we saw in section 1.2, line and bar graphs are useful for showing frequencies of data values. When a dataset contains many distinct values, such as the lamp lifetimes in Table 1.3, creating a frequency table for each individual value becomes impractical. A more effective approach is to group the data into a set of **class intervals**, which are also commonly referred to as **bins**.

Item Lifetimes (in Hours)									
1067	919	1196	785	1126	936	918	1156	920	948
855	1092	1162	1170	929	950	905	972	1035	1045
1157	1195	1195	1340	1122	938	970	1237	956	1102
1022	978	832	1009	1157	1151	1009	765	958	902
923	1333	811	1217	1085	896	958	1311	1037	702
521	933	928	1153	946	858	1071	1069	830	1063
930	807	954	1063	1002	909	1077	1021	1062	1157
999	932	1035	944	1049	940	1122	1115	833	1320
901	1324	818	1250	1203	1078	890	1303	1011	1102
996	780	900	1106	704	621	854	1178	1138	951
1187	1067	1118	1037	958	760	1101	949	992	966
824	653	980	935	878	934	910	1058	730	980
844	814	1103	1000	788	1143	935	1069	1170	1067
1037	1151	863	990	1035	1112	931	970	932	904
1026	1147	883	867	990	1258	1192	922	1150	1091
1039	1083	1040	1289	699	1083	880	1029	658	912
1023	984	856	924	801	1122	1292	1116	880	1173
1134	932	938	1078	1180	1106	1184	954	824	529
998	996	1133	765	775	1105	1081	1171	705	1425
610	916	1001	895	709	860	1110	1149	972	1002

Table 1.3: Life in Hours of 200 Incandescent Lamps.

The choice of how many bins to use involves a trade-off. If there are too few bins, we risk losing important

information by grouping too many distinct values together. If there are too many bins, the frequencies within each bin may become too small to reveal any clear pattern in the data's distribution.

While the optimal number of bins is a subjective choice that depends on the dataset, a selection of 5 to 10 bins is typical. It is also customary, though not required, to use bins of equal width, as this often makes interpretation easier.

The endpoints of a bin are called its **class boundaries**. We will adopt the common **left-end inclusion convention**, which specifies that a bin contains its left-end boundary but not its right-end one. For example, the bin 700–800 would contain any value x such that $700 \leq x < 800$.

Following this approach, the raw data from Table 1.3 is summarized in Table 1.4 using bins of length 100, starting at 500.

Class Interval	Frequency	Relative Freq.	Cumulative Rel. Freq.
500–600	2	0.010	0.010
600–700	5	0.025	0.035
700–800	12	0.060	0.095
800–900	25	0.125	0.220
900–1000	58	0.290	0.510
1000–1100	41	0.205	0.715
1100–1200	43	0.215	0.930
1200–1300	7	0.035	0.965
1300–1400	6	0.030	0.995
1400–1500	1	0.005	1.000

Table 1.4: Frequency Distribution for the Lifetimes of 200 Incandescent Lamps.

A bar graph that plots the frequencies of data grouped into bins is called a **histogram**. A key feature of a histogram is that the bars are placed directly adjacent to one another, reflecting the continuous nature of the bins. The vertical axis can represent either the absolute frequency (the count of values in each interval) or the relative frequency. The former is called a **frequency histogram**, while the latter is a **relative frequency histogram**. Figure 1.4 shows a frequency histogram for the lamp lifetime data.

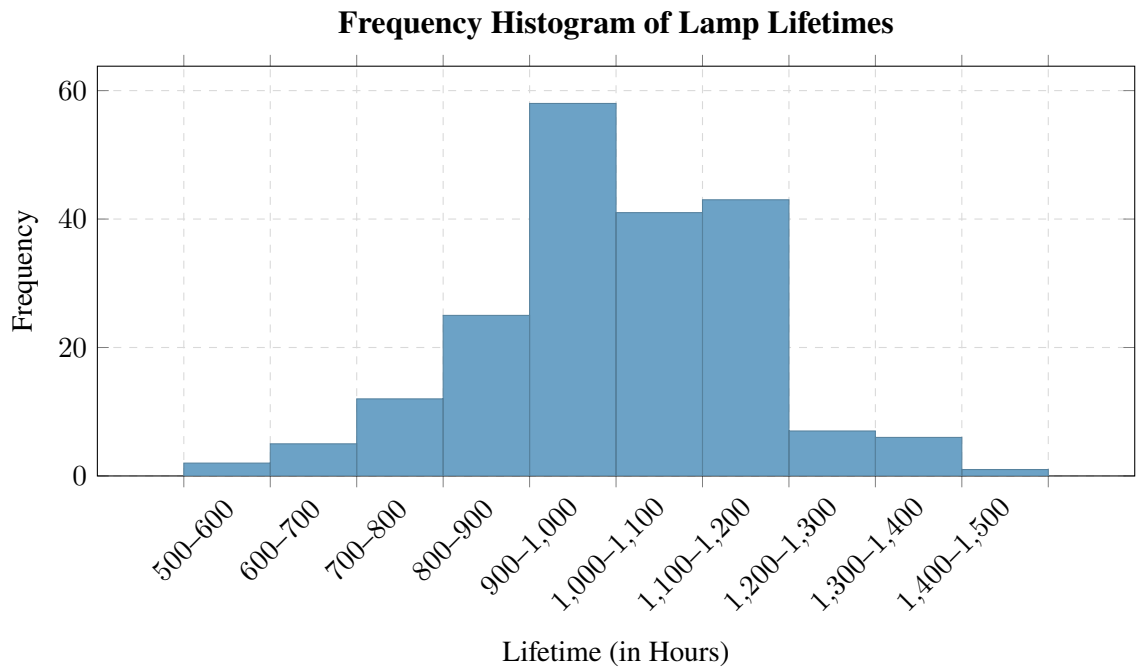


Figure 1.4: A frequency histogram for the lamp lifetime data. The adjacent bars show the distribution of data across continuous intervals.

Finally, we are sometimes interested in plotting the cumulative frequency or, more commonly, the cumulative relative frequency. Such a graph, often called an **ogive**, shows the number or proportion of data points that fall at or below a certain value.

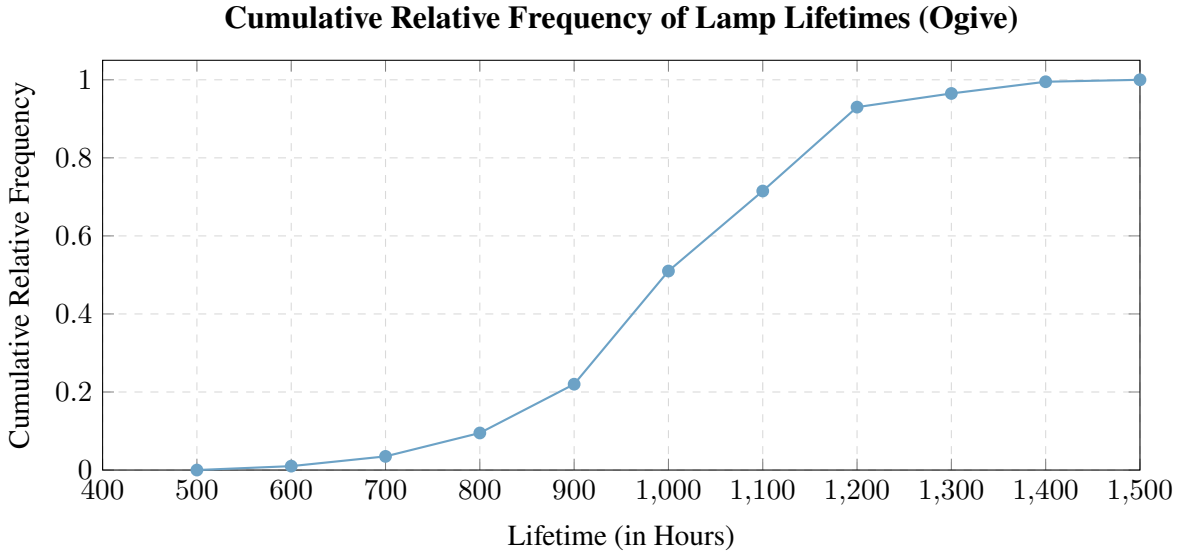


Figure 1.5: An ogive showing the cumulative relative frequency for the lamp lifetime data.

On an ogive, each point on the horizontal axis represents a data value, while the corresponding point on the vertical axis shows the proportion of the data that is less than or equal to that value. For example, Figure 1.5 presents the cumulative relative frequency plot for the lamp lifetime data. From this graph, we can quickly determine that approximately 50% of the lamps had a lifetime of less than 1000 hours, about 95% had a lifetime of less than 1200 hours (the actual value is 93%), and 100% of the lamps failed by 1500 hours.

1.3 Summarizing Data

Having seen how to visually and numerically summarize data distributions, we now focus on two fundamental aspects of describing a dataset: its *center* (or typical value) and its *dispersion* (or variability).

Measures of Central Tendency

The **mean**, often called the average, is the most commonly used measure of central tendency. Definition 1.6 shows how to calculate the mean of a dataset and Example 1.4 exemplifies the calculation.

Definition 1.6 (Arithmetic Mean)

For a dataset with n values $x_1, x_2, \ldots, x_n$, the **arithmetic mean** is defined as:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

♣

Example 1.4 Response Time Analysis

A web application recorded the following response times (in milliseconds) for 10 requests:

$$\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$$

The mean response time is:

$$\bar{x} = \frac{120 + 150 + 180 + 200 + 220 + 250 + 280 + 300 + 350 + 400}{10} = \frac{2450}{10} = 245 \text{ milliseconds}$$

The mean incorporates every value in a dataset, which makes it a powerful measure of central tendency. It is most appropriate when the data values are of similar magnitude and when the distribution is reasonably symmetric without extreme outliers. Under these conditions, the mean provides a reliable summary of the dataset.

For example, in the analysis of web performance, the mean response time offers an overall indication of how quickly a system responds on average. However, if a few response times are much slower than the rest, the mean may give a misleading impression of the typical user experience.

Beyond its role as a summary statistic, the mean also serves as a foundation for many statistical methods. In particular, it is central to the definitions of variance and standard deviation, which measure the dispersion of data around the mean.

Remark: As mentioned earlier, and as we will revisit, the mean is highly sensitive to extreme values or outliers. Even a single unusually large or small observation can shift the mean considerably, reducing its ability to reflect the typical value of the dataset. It is therefore important to consider the distribution of the data before relying solely on the mean to describe the center of a dataset.

The **median** is the middle value when data is arranged in ascending order. It is less sensitive to outliers than the mean.

Definition 1.7 (Median)

For a dataset with n values arranged in ascending order, the median is defined as:

$$\text{median} = \begin{cases} x_{\frac{n+1}{2}} & \text{if } n \text{ is odd} \\ \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2} & \text{if } n \text{ is even} \end{cases}$$

♣

Example 1.5 Median Response Time

Using the same response time data from the previous example:

$$\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$$

Since $n = 10$ (even), the median is the average of the 5th and 6th values:

$$\text{Median} = \frac{220 + 250}{2} = 235 \text{ milliseconds}$$

Example 1.6

Consider the same data with an extreme value:

$$\{120, 150, 180, 200, 220, 250, 280, 300, 350, 2000\}$$

The mean becomes: $\bar{x} = \frac{4050}{10} = 405$ milliseconds (significantly affected by the outlier)

The median remains: $\frac{220+250}{2} = 235$ milliseconds (unaffected by the outlier)

Because the median divides a dataset into two equal halves, it serves as a robust measure of central tendency, particularly when the data contain outliers or are skewed. Unlike the mean, which can be drawn toward extreme values, the median remains largely unaffected, offering a more reliable indication of the "typical" value in such situations.

The **mode** is the most frequently occurring value in a dataset.

Definition 1.8 (Mode)

The **mode** is the value that appears most frequently in a dataset. A dataset can have:

- **No mode**: If all values appear with equal frequency
- **One mode (unimodal)**: If one value appears most frequently
- **Multiple modes (multimodal)**: If two or more values tie for the highest frequency

♣

Sometimes, when a data set has two modes, it is called a **bimodal** dataset.

Example 1.7

A software project has the following bug severity levels:

$$\{\text{High}, \text{Medium}, \text{Low}, \text{High}, \text{Critical}, \text{Medium}, \text{High}, \text{Low}, \text{High}, \text{Medium}\}$$

Counting frequencies:

- High: 4 occurrences
- Medium: 3 occurrences
- Low: 2 occurrences
- Critical: 1 occurrence

The mode is ”High” since it appears most frequently.

The mode is especially useful when analyzing categorical or discrete data, where calculating a mean or median may not make sense. For example, in software engineering, the mode can help identify the most common error code returned by an API, the most frequently used programming language in a codebase, or the most reported type of bug in an issue tracker.

When to Use Each Measure

Measure	Best For	Limitations
Mean	Continuous data, symmetric distributions	Sensitive to outliers
Median	Data with outliers, skewed distributions	Less informative for symmetric data
Mode	Categorical data, discrete data	May not exist or be unique

Table 1.5: Comparison of Central Tendency Measures

Measures of Dispersion

While measures of central tendency tell us about the typical value, **measures of dispersion** describe how spread out the data are.Understanding variability is crucial in software engineering for assessing consistency and reliability.

The **range** is the simplest measure of dispersion.

Definition 1.9 (Range)

The **range** of a dataset is the difference between the maximum and minimum values:

$$\text{Range} = x_{\max} - x_{\min}$$

♣

Example 1.8

For the response time data: $\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$

$$\text{Range} = 400 - 120 = 280 \text{ milliseconds}$$

Remark: The range is sensitive to outliers and doesn't provide information about the distribution of values between the extremes.

The **variance** and **standard deviation** are more sophisticated measures that consider all data points.

Definition 1.10 (Sample Variance and Standard Deviation)

For a sample with n values, the **sample variance**, denoted by s^2, is defined as:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

The **sample standard deviation**, denoted by s, is the square root of the variance:

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$



While the mean identifies the center of a dataset, the **variance** and **standard deviation** measure its variability or spread. A small standard deviation indicates that data points are clustered tightly around the mean, whereas a large standard deviation signifies that they are more spread out.

The sample variance is a scaled sum of the squared deviations from the mean, using the divisor $n-1$. Squaring each deviation serves two purposes: it makes all contributions non-negative, and it gives greater weight to larger deviations. However, this calculation leaves the variance in squared units (e.g., milliseconds squared), which are not intuitive to interpret.

To solve this, we use the standard deviation, which is simply the square root of the variance. This crucial step returns the measure to the original units of the data (e.g., milliseconds), making it directly interpretable. In short, the standard deviation represents the typical distance of a data point from the mean.

Example 1.9 Calculating Standard Deviation

For the response time data: $\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$ with $\bar{x} = 245$:

$$\begin{aligned}
s^2 &= \frac{1}{9}[(120-245)^2 + (150-245)^2 + \cdots + (400-245)^2] \\
&= \frac{1}{9}[(-125)^2 + (-95)^2 + \cdots + (155)^2] \\
&= \frac{1}{9}[15625 + 9025 + \cdots + 24025] \\
&= \frac{1}{9} \times 70850 \approx 7872.22
\end{aligned}$$

$$s = \sqrt{7872.22} \approx 88.73 \text{ milliseconds}$$

The following algebraic identity is often helpful when computing variance or standard deviation by hand:

We start from the definition:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Using the identity

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n}x_i^2 - \frac{\left(\sum_{i=1}^{n}x_i\right)^2}{n}$$

we can rewrite sample variance as

$$s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}\right) = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right)$$

This is known as the **computational formula for variance**, as it allows you to calculate s^2 directly from $\sum x_i$ and $\sum x_i^2$ (or equivalently, from $\sum x_i^2$ and $\bar{x}$), without needing to compute each individual deviation from the mean.

The **coefficient of variation** allows comparison of variability across datasets with different scales.

Definition 1.11 (Coefficient of Variation)

The **coefficient of variation** is the ratio of standard deviation to the mean:

$$CV = \frac{s}{\bar{x}} \times 100\%$$

♣

Example 1.10 Comparing Variability

Consider two systems:

- System A: Mean response time = 100ms, Standard deviation = 20ms
- System B: Mean response time = 500ms, Standard deviation = 100ms

$$CV_A = \frac{20}{100} \times 100\% = 20\%$$
$$CV_B = \frac{100}{500} \times 100\% = 20\%$$

Both systems have the same relative variability (20%), despite different absolute standard deviations.

Measures of Position

To introduce a measure of spread that is less sensitive to outliers, we first need to understand measures of position, which describe the relative standing of a data point within a dataset.

Definition 1.12 (Percentile)

The p-th **percentile** is a value at or below which approximately $p\%$ of the observations fall. Several conventions are used to calculate percentiles, especially for small samples, so the chosen method should always be stated.

♣

For example, if a response time is at the 95th percentile, it means that 95% of all response times are faster than this one.

Example 1.11 Percentile Calculation

For the response time data, use the nearest-rank method to find the 90th percentile:

1. Sort data: $\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$
2. Rank: $\lceil 0.9 \times 10 \rceil = 9$
3. 90th percentile = 350ms

Quartiles are specific, widely used percentiles that divide the data into four equal parts.

Definition 1.13 (Quartiles)

Quartiles divide an ordered dataset into four equal parts.

- **Q1 (First Quartile)**: The 25th percentile.
- **Q2 (Second Quartile)**: The 50th percentile. This is the **median** of the dataset.
- **Q3 (Third Quartile)**: The 75th percentile.

♣

Example 1.12 Calculating Quartiles

For the sorted response time data: $\{120, 150, 180, 200, 220, |250, 280, 300, 350, 400\}$

- The lower half is $\{120, 150, 180, 200, 220\}$. The median of this half is $Q1 = 180$ ms.
- The upper half is $\{250, 280, 300, 350, 400\}$. The median of this half is $Q3 = 300$ ms.

Example 1.13 Quartiles from an Ogive

Quartiles can be read directly from an ogive. As an example, we will again use the lamp lifetime data. Reading from the ogive in Figure 1.6, the approximate quartiles are $Q1 \approx 900$, $Q2$ (median) ≈ 1000, and $Q3 \approx 1100$ hours.

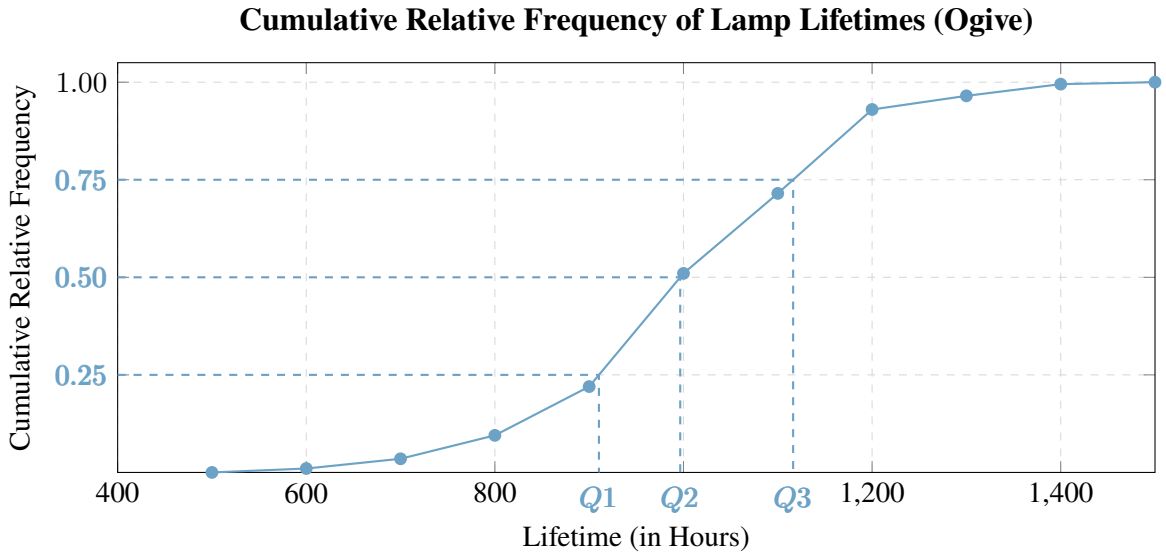


Figure 1.6: An ogive for the lamp lifetime data with quartiles $Q1$, $Q2$, and $Q3$ indicated by dashed lines.

Percentiles divide data into 100 equal parts, while quartiles divide data into four equal parts.

The **interquartile range** is a robust measure of dispersion that is generally much less sensitive to extreme observations than the full range or the standard deviation.

Definition 1.14 (Interquartile Range)

The **interquartile range (IQR)** measures the spread of the middle 50% of the data and is defined as

$$IQR = Q_3 - Q_1,$$

where Q_1 and Q_3 are the first and third quartiles. A larger IQR indicates that the central portion of the

data is more widely dispersed, while a smaller IQR indicates that it is more tightly clustered. ♣

Example 1.14 IQR Calculation

For the response time data: $\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$

- Lower half: $\{120, 150, 180, 200, 220\} \rightarrow Q1 = 180$
- Upper half: $\{250, 280, 300, 350, 400\} \rightarrow Q3 = 300$
- $IQR = 300 - 180 = 120$ milliseconds

Summary Statistics

Before we conclude this chapter, we will review the summary statistics covered in this section. Table 1.6 is a hypothetical example for a separate dataset of 1000 web response times. The underlying observations are not listed, so the values are included to illustrate the structure of a summary table rather than as a reproducible calculation from the preceding ten observations.

Example 1.15

For a dataset of 1000 web response times:

Measure	Value
Count	1000
Mean	245.3 ms
Median	238.0 ms
Mode	220.0 ms
Standard Deviation	95.6 ms
Variance	9139.4 ms^2
Minimum	45.0 ms
Maximum	1200.0 ms
Range	1155.0 ms
Q1	180.0 ms
Q3	300.0 ms
IQR	120.0 ms
Skewness	1.2 (positive skew)
Kurtosis	2.8 (leptokurtic)

Table 1.6: Complete Statistical Summary

This table provides a complete picture of the data and is often a first step in the analysis of a dataset. Note that the skewness and kurtosis have not been discussed but will be explained in the next section. We have chosen to include them here for completeness as they are often used in the analysis of a dataset - although how to calculate them is beyond the scope of this book.

1.4 Understanding Data Distributions

Once we have visualized our data with histograms and box plots, we can begin to analyze the *shape* of the distribution. The shape tells us about the underlying patterns in our data and helps us choose the right statistical tools. In software engineering, understanding the distribution of metrics like response times or error rates is critical for setting performance baselines and detecting anomalies. First we review the concept of outliers.

Normal Data Sets

Many large data sets yield histograms with a distinctive overall shape: the frequencies peak near the sample median and decrease symmetrically on both sides, forming a characteristic **bell-shaped curve**. Data sets exhibiting this property are called **normal**, and their histograms are referred to as **normal histograms** (see Figure 1.7).

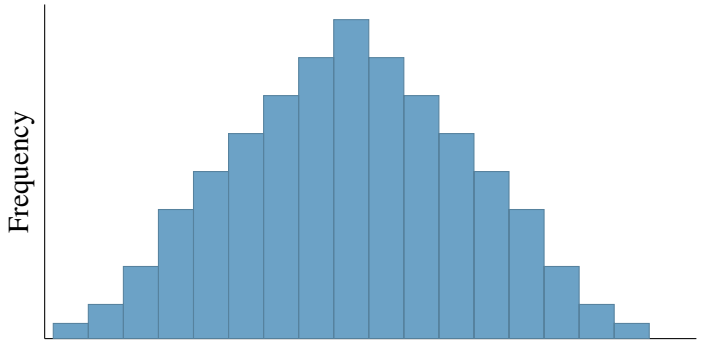


Figure 1.7: A histogram of a normal data set, showing the characteristic bell shape.

In software systems, perfectly normal distributions are rare, but some metrics can be **approximately normal**. For example, the latency of a highly optimized, internal microservice that performs a consistent task might be approximately normal. A histogram of an approximately normal data set looks like Figure 1.8.

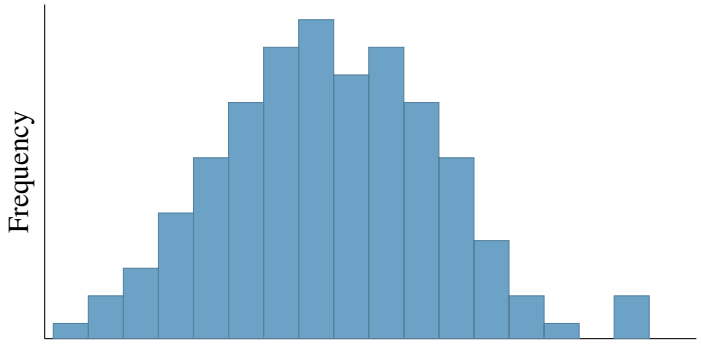


Figure 1.8: A histogram of a data set that is approximately normal. While not perfectly symmetric, it follows the general bell shape.

Many datasets observed in practice, from human height to measurement errors, follow this specific shape. Such data is said to be **normally distributed**.

Definition 1.15 (Normal Distribution)

A **normal distribution** is a symmetric, bell-shaped distribution where the mean, median, and mode are all equal and located at the center. Its shape is determined entirely by its mean (μ) and standard deviation (σ).

A histogram estimates the shape of the distribution that generated the observations. With more data and an appropriate choice of bin width, it can give a more detailed picture of the underlying distribution, but it does not necessarily become normal. Whether a normal model is appropriate depends on the data-generating process.

The mathematical normal model, its density function, and probability calculations are developed in chapter 4. Its probability density has the following characteristic shape:

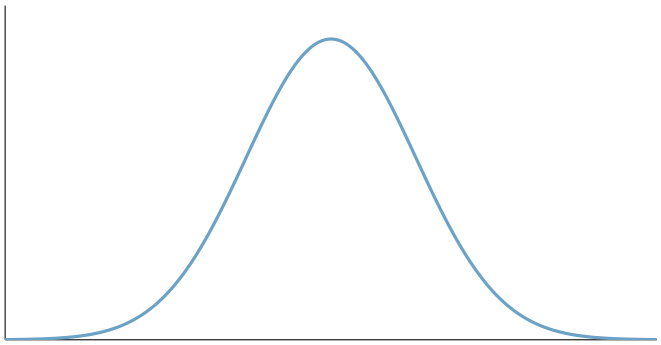


Figure 1.9: The normal distribution curve.

Skewed Data Sets

When a data set is not symmetric about its sample median, it is described as **skewed**. A long tail extending to the right indicates the distribution is **skewed to the right**; a long tail to the left means it is **skewed to the left**. Thus, the histogram in Figure 1.10 is skewed left, while the one in Figure 1.11 is skewed right.

Definition 1.16 (Skewness)

Skewness measures the asymmetry of data distribution:

- **Positive skew (skewed to the right)**: Tail extends to the right (mean ¿ median)
- **Negative skew (skewed to the left)**: Tail extends to the left (mean ¡ median)
- **Symmetric**: Mean ≈ median

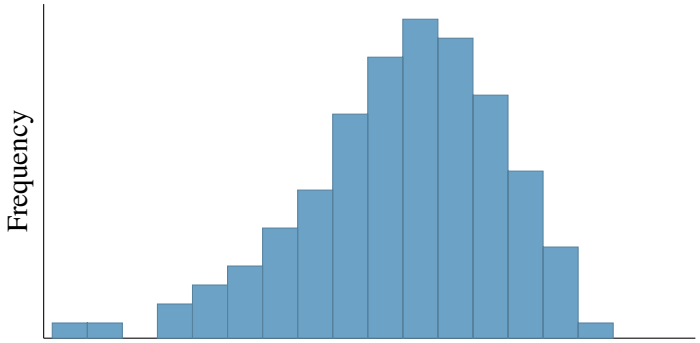


Figure 1.10: A histogram of a data set that is skewed to the left (negatively skewed). The tail of the distribution is on the left.

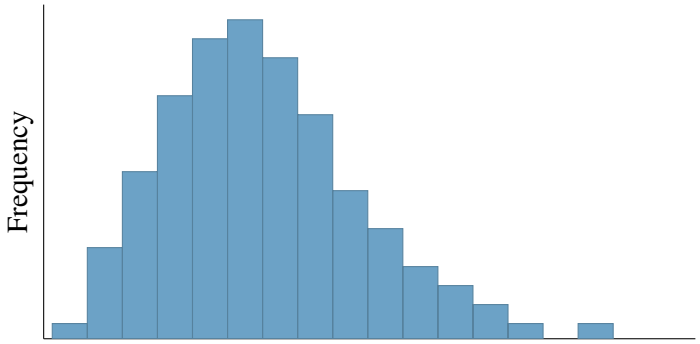


Figure 1.11: A histogram of a data set that is skewed to the right (positively skewed). The tail of the distribution is on the right.

Figure 1.12 illustrates common patterns for the mean, median, and mode in unimodal skewed distributions. These orderings are useful heuristics, not definitions of skewness and not universal rules.

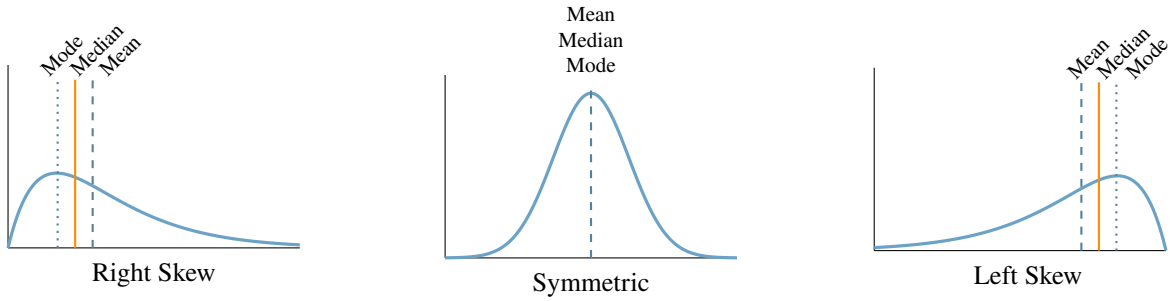


Figure 1.12: Relationship between mean, median, and mode for positively skewed, symmetric, and negatively skewed distributions.

Outlier Detection

As mentioned in section 1.1, outliers are data points that are significantly different from the rest of the data. They can arise due to measurement errors, data entry errors, or genuine variability in the data. Detecting outliers is important because they can distort statistical analyses.

A common method for identifying outliers is to use the interquartile range (IQR):

- **Mild outliers**: Data points outside the $1.5 \times IQR$ fences but not outside the $3 \times IQR$ fences.
- **Extreme outliers**: Data points less than $Q_1 - 3 \times IQR$ or greater than $Q_3 + 3 \times IQR$.

Example 1.16 Detecting Outliers

Given the response time data: $\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$

- $Q_1 = 180$, $Q_3 = 300$, $IQR = 120$
- Lower fence: $Q_1 - 1.5 \times IQR = 180 - 180 = 0$
- Upper fence: $Q_3 + 1.5 \times IQR = 300 + 180 = 480$

All data points are between 0 and 480, so there are no mild outliers in this dataset.

Example 1.17

Suppose we have the following set of response times (in milliseconds) for a web application:

$\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400, 1200\}$.

Let's identify if there is an upper outlier:

- Sorted data: $120, 150, 180, 200, 220, 250, 280, 300, 350, 400, 1200$
- $n = 11$
- Median (Q_2): 250 (6th value)
- Lower half: $120, 150, 180, 200, 220 \rightarrow Q_1 = 180$
- Upper half: $280, 300, 350, 400, 1200 \rightarrow Q_3 = 350$
- $IQR = Q_3 - Q_1 = 350 - 180 = 170$
- Upper fence: $Q_3 + 1.5 \times IQR = 350 + 1.5 \times 170 = 350 + 255 = 605$

The value 1200 is greater than 605, so it is an upper outlier.

Box Plots

Box plots provide a compact summary of data distribution, highlighting the median, quartiles, and outliers. Figure 1.13 shows an example of a box plot.

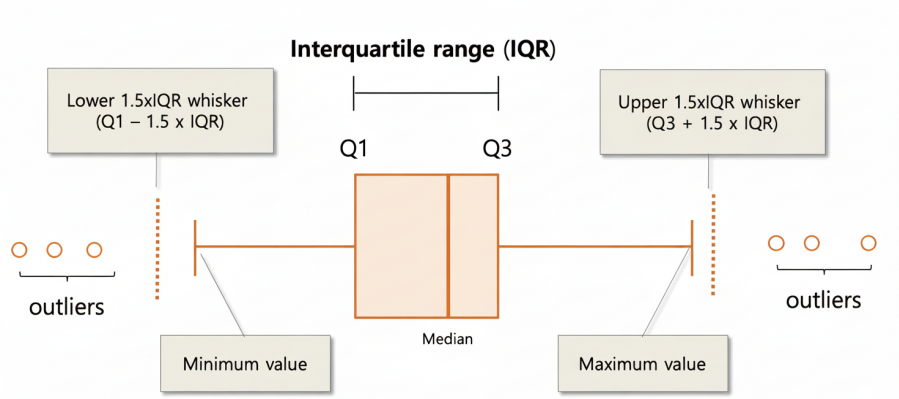


Figure 1.13: An example of a box plot, illustrating the median, quartiles, whiskers, and potential outliers.

Definition 1.17 (Box Plot Components)

A box plot is based on the five-number summary and may also display outliers:

- **Lower whisker**: Usually the smallest observation within the lower $1.5 \times IQR$ fence
- **Q1 (First Quartile)**: The 25th percentile
- **Median (Q2)**: The 50th percentile
- **Q3 (Third Quartile)**: The 75th percentile
- **Upper whisker**: Usually the largest observation within the upper $1.5 \times IQR$ fence
- **Outliers**: Points beyond $1.5 \times IQR$ from the box

A box plot provides a visual summary of the distribution of a dataset, making it easy to compare groups and spot outliers. Here's how to read a box plot.

- The **length of the box** shows the spread of the central half of the data. A longer box means more variability in the middle 50%.
- The **position of the median line** and the relative whisker lengths can suggest asymmetry, but a box plot alone does not determine the direction or degree of skewness.
- In many unimodal skewed distributions, the mean is pulled toward the longer tail. Consequently, one often sees mode $<$ median $<$ mean for right-skewed data and the reverse ordering for left-skewed data. These are common patterns rather than guaranteed identities.
- The **whiskers** show the range of the bulk of the data, excluding outliers.
- **Outliers** highlight unusually high or low values that may warrant further investigation.

Box plots are especially useful for comparing distributions across several groups or datasets, as they quickly reveal differences in medians, spreads, and the presence of outliers.

This concludes the foundational treatment of data visualization and descriptive summary statistics. The next chapter develops the probability concepts needed for statistical inference.

Chapter 2 Probability for Data Analysis

Probability provides a language for reasoning when the available information is incomplete. In engineering and data analysis, uncertainty appears whenever measurements vary, components can fail, users behave differently, or observations reveal only part of an underlying process. The purpose of a probability model is not to eliminate this uncertainty, but to describe it precisely enough that conclusions and decisions can be justified.

A famous illustration is the Monty Hall problem. Imagine a game show with three closed doors. A car is hidden behind one door and a goat behind each of the other two. You choose a door, but before it is opened, the host—who knows where the car is and always opens a door hiding a goat—opens one of the remaining doors. You are then offered the choice of staying with your original door or switching to the other unopened door. At first sight, the two unopened doors may appear equally likely. After all, only two possibilities remain.

The crucial point is that the host's action is not a random observation made without knowledge. The door that the host opens depends on both the location of the car and the door initially selected by the contestant. Consequently, the information supplied by the host must be interpreted conditionally. A careful probability model shows that the original choice still has probability (1/3) of hiding the car, while the switching strategy wins with probability (2/3). The puzzle is surprising because intuitive reasoning often ignores the mechanism that produced the observed information.

This is exactly the kind of problem for which conditional probability and Bayes' theorem are designed. Bayes' theorem allows probabilities to be revised when new evidence becomes available while taking account of how likely that evidence would be under each possible explanation. The same reasoning appears in diagnostic testing, fault detection, quality control, spam filtering, and many other data-driven systems. We build toward Bayes' theorem gradually: beginning with random experiments, sample spaces, and events, then developing the basic probability rules, conditional probability, total probability, and independence. The chapter concludes with Bayes' theorem in section 2.7.

2.1 Random Experiments, Sample Spaces, and Events

A **random experiment** is one that can lead to different outcomes, even when repeated under the same conditions. This randomness is a fundamental aspect of many engineering tasks.

Think of it this way: Let us say you are testing the speed of a website under different conditions. Sometimes it loads quickly, and other times it is slower. Even if you are using the same code and server, things like network traffic or server load make the results vary each time.

Definition 2.1 (Random Experiment)

A random experiment is one that can give different results, even if you do everything the same each time. ♣

Or, imagine you are measuring the signal strength in a wireless device. You might get slightly different readings each time because of things like interference or small changes in the environment.

This randomness shows up all over the place, from software performance tests to electrical engineering experiments. It is important to expect it and include it in your thinking. Otherwise, you might make decisions based

on incomplete or misleading data. When you account for random variation, you can make smarter predictions and designs.

To model and analyze a random experiment, it is crucial to understand the set of possible outcomes that can occur. In probability theory, this set is called the **sample space**, denoted by S. A sample space can be either **discrete** (consisting of a finite or countably infinite set of outcomes) or **continuous** (containing an interval of real numbers). The exact definition of a sample space often depends on the objectives of the analysis.

An **outcome** is a single possible result of the random experiment, and an **event** is any subset of the sample space, which may consist of one or more outcomes. Below are some examples to illustrate these concepts:

Example 2.1 Network Latency

Consider an experiment where you measure the latency of data packets in a network. The sample space can be defined based on the type of measurements:

- If latency is measured as a positive real number, the sample space is continuous:
  $S = \{x \mid x > 0\}$.
- If it is known that latency ranges between 10 and 100 milliseconds, the sample space can be refined to:
  $S = \{x \mid 10 \leq x \leq 100\}$.
- If the objective is to categorize latency as low, medium, or high, the sample space becomes discrete:
  $S = \{\text{low}, \text{medium}, \text{high}\}$.
- For a simple evaluation of whether the latency meets a standard threshold, the sample space can be reduced to:
  $S = \{\text{pass}, \text{fail}\}$.

Each outcome in these sample spaces represents a single possible latency measurement, and events can be defined as sets of outcomes, such as "latency is high."

Understanding the nature of sample spaces, outcomes, and events is fundamental in probability, as it allows us to define and work with probabilities of complex scenarios in various engineering contexts. Let us summarise these key concepts:

Definition 2.2 (Sample Spaces, Outcomes, and Events)

Sample Space: The set of all possible outcomes of a random experiment is called the sample space, denoted by S. Outcomes can be discrete or continuous, depending on the nature of the experiment.

Outcome: A single possible result of a random experiment.

Event: Any subset of the sample space, which may consist of one or more outcomes. ♣

Example 2.2 Software Release Testing

Imagine a software testing process where each test case can either pass or fail. The sample space for a single test case is discrete and can be represented as:

$$S = \{\text{pass}, \text{fail}\}.$$

If you run three test cases, the combined sample space for all possible outcomes is:

$$S = \{(\text{pass, pass, pass}), (\text{pass, pass, fail}), (\text{pass, fail, pass}),$$
$$(\text{pass, fail, fail}), (\text{fail, pass, pass}), (\text{fail, pass, fail}),$$
$$(\text{fail, fail, pass}), (\text{fail, fail, fail})\}.$$

This sample space includes all sequences of outcomes for the three tests.

- The total number of possible outcomes is $2^3 = 8$.
- An event could be defined as "at least one test fails," which would include outcomes like (fail, pass, pass), (pass, fail, fail), and others where at least one test fails.

Example 2.3 Component Quality in Manufacturing

A company manufactures electronic components, and each component is tested for compliance with quality standards. The test can return one of three outcomes: `pass`, `marginal`, or `fail`. The sample space is:

$$S = \{\text{pass}, \text{marginal}, \text{fail}\}.$$

Event: Suppose we are interested in the event that a component does not pass the quality test. This event is a set of outcomes:

$$E = \{\text{marginal}, \text{fail}\}.$$

This means that all the operations we have defined on sets translate directly into operations on events:

- $A \cup B$: the event that at least one of A or B occurs.
- $A \cap B$: the event that both A and B occur together.
- $\overline{A}$ or A^c: the event that A does not occur.
- $A \setminus B$ or $A - B$: the event that A occurs but B does not.
- $A \Delta B$ or $A \oplus B$: the event that either A or B occurs, but not both (symmetric difference).

Thus, probability theory builds directly on set theory: probability assigns a numerical measure to these subsets of the sample space. Later in this chapter, we will see how this measure is defined and used to reason about uncertainty.

We can summarise the considerations of this section in the following table:

Operation	Boolean Algebra	Logic	Set Theory
`NOT`	$\overline{x}$	$\neg x$	A^c or A'
`OR`	$+$	$\vee$	$\cup$
`AND`	$\cdot$	$\wedge$	$\cap$
`NAND`	$\overline{x \cdot y}$	$\neg(x \wedge y)$	$(A \cap B)^c$ or $\overline{A \cap B}$
`NOR`	$\overline{x + y}$	$\neg(x \vee y)$	$(A \cup B)^c$ or $\overline{A \cup B}$
`XOR` (Symmetric Difference)	$x \oplus y$	$(x \wedge \neg y) \vee (\neg x \wedge y)$	$A \Delta B$ or $A \oplus B$
Difference	$x \cdot \overline{y}$	$x \wedge \neg y$	$A - B$ or $A \setminus B$

Table 2.1: Comparison of Operators in Boolean Algebra, Logic, and Set Theory

In this book, we will use the notation A^c to denote the complement of the set A, which includes all elements not in A.

2.2 Basic Probability

Probability quantifies the likelihood or chance of events associated with a random experiment. For instance, when you hear, "The chance of rain today is 30%," it expresses uncertainty about the event that it rains. Probabilities range from 0 to 1 (or equivalently, from 0% to 100%). An impossible event has probability 0, and the entire sample space has probability 1. In general probability models, however, an event can have probability 0 without being empty, or probability 1 without containing every possible outcome; such events are called almost impossible and almost sure, respectively.

Probabilities can be interpreted in different ways:

- **Objective (or Classical) Probability**: Often referred to as classical probability, this approach is used when outcomes are equally likely, such as in rolling a fair die or flipping a coin. Probabilities are assigned based on the assumption that each outcome has an equal chance of occurring. For example, when rolling a fair six-sided die, the probability of rolling a 3 is $\frac{1}{6}$ because there are 6 equally likely outcomes (1, 2, 3, 4, 5, 6), and only one of them is a 3. The probability is the same for all observers.
- **Relative Frequency (Empirical Probability)**: Empirical probability is based on observations from experiments rather than theoretical calculations. For example, if a software tester runs a stress test on a server 100 times, and it crashes 7 times, the empirical probability of a crash is $\frac{7}{100} = 0.07$. This approach relies on actual data rather than assumptions or intuition.
- **Subjective Probability**: This reflects our personal belief or degree of confidence in an outcome. Different people might assign different probabilities to the same event based on their knowledge or perspective. You and your friends discuss Denmark's chances of winning the World Cup. Based on recent performance and team strength, you estimate a 10% chance. However, a more optimistic friend assigns a 20% chance, while another gives only 5%, considering stronger competitors. This illustrates subjective probability, where each person's estimate varies based on personal beliefs and biases rather than objective data.

When assigning probabilities, it's essential that the sum of all probabilities in an experiment equals 1, ensuring consistency with the relative frequency interpretation.

We start by establishing the Axioms of Probability, which lay the foundation for how probabilities are assigned to events. These axioms define the basic properties that every probability measure must satisfy.

Axiom 2.1 (Axioms of Probability)

- **Axiom 1:** For any event A, $0 \leq P(A) \leq 1$.
- **Axiom 2:** Probability of the sample space S is $P(S) = 1$.
- **Axiom 3:** If $A_1, A_2, A_3, \cdots$ are disjoint events, then $P\left(A_1 \cup A_2 \cup A_3 \cdots\right) = P\left(A_1\right) + P\left(A_2\right) + P\left(A_3\right) + \cdots$

♡

The property that $0 \leq P(A) \leq 1$ is equivalent to the requirement that a relative frequency must be between 0 and 1. The property that $P(S) = 1$ is a consequence of the fact that an outcome from the sample space occurs on every trial of an experiment. Consequently, the relative frequency of S is 1. Property 3 implies that if the events A_1 and A_2 have no outcomes in common, the relative frequency of outcomes in $A_1 \cup A_2$ is the sum of

the relative frequencies of the outcomes in A_1 and A_2.

In the next sections we will see more about the probability of events and how to calculate them.

Probability of an Event

The probability of an event is a measure of the likelihood that the event will occur. It is denoted by $P(A)$, where A is the event, and it ranges from 0 to 1. The empty event has probability 0, and the sample space has probability 1.

Definition 2.3 (Classical Probability for Equally Likely Outcomes)

If a finite sample space consists of equally likely outcomes, the probability of an event A is the ratio of the number of outcomes in A to the total number of outcomes in the sample space:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$



Example 2.4 Suppose you are testing a software module with 10 different test cases. Out of these, 3 test cases are known to fail due to a bug. If you randomly select one test case to run, what is the probability that the selected test case will fail?

Solution: Here, the event A is "the test case fails."

- Number of favorable outcomes (failing test cases) $= 3$
- Total number of outcomes (total test cases) $= 10$

Using the formula:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} = \frac{3}{10} = 0.3$$

Therefore, the probability that a randomly selected test case will fail is 0.3 , indicating that there is a 30% chance of failure. ◀

Example 2.5 Imagine a software development environment where you have 50 files, consisting of 20 JavaScript files, 15 Java files, and 15 configuration files. If you randomly select one file to edit, what is the probability that the file is a JavaScript file?

Solution: Here, the event A is "the selected file is a JavaScript file."

- Number of favorable outcomes (JavaScript files) $= 20$
- Total number of outcomes (total files) $= 50$

Using the formula:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} = \frac{20}{50} = 0.4$$

Thus, the probability of selecting a JavaScript file is 0.4 , meaning there is a 40% chance of choosing a JavaScript file from the set. ◀

2.3 Probability of Joint Events and Set Operations

Joint events are formed by applying basic set operations to individual events. Commonly, we encounter unions of events, such as $A \cup B$; intersections of events, such as $A \cap B$; and complements of events, such as A^c. These combined events are often of particular interest, and their probabilities can frequently be derived from the probabilities of the individual events that compose them. Understanding these set operations is essential for accurately calculating the probability of joint events. In this section, we will explore how unions of events and other set operations can be used to determine the probabilities of more complex events.

When dealing with events, the intersection represents `AND` while the union represents `OR` The probability of the intersection of events A and B, denoted as $P(A \cap B)$, can also be expressed as $P(A, B)$ or $P(AB)$.

From the axioms of probability, we can derive the following rules of probabilities:

Theorem 2.1 (Rules of Probability)

- **Complement Rule:** The probability of the complement of event A is
  $$P(A^c) = 1 - P(A)$$
- **Empty Set Rule:** The probability of the empty set is 0, i.e.,
  $$P(\emptyset) = 0$$
- **Addition Rule:** For any two events A and B, the probability of the union of events A and B is given by
  $$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
- **Difference Rule:** The probability of the difference between events A and B is given by
  $$P(A - B) = P(A) - P(A \cap B)$$
- **Subset Rule:** If A is a subset of B ($A \subset B$), then
  $$P(A) \leq P(B)$$

.



We can obtain the Complement Rule by noting:

$$\begin{aligned} 1 &= P(S) && \text{(axiom 2)} \\ &= P(A \cup A^c) && \text{(definition of complement)} \\ &= P(A) + P(A^c) && \text{(since } A \text{ and } A^c \text{ are disjoint)} \end{aligned}$$

Since $\emptyset = S^c$, we can apply part the Complement Rule to deduce that $P(\emptyset) = 1 - P(S) = 0$. This is intuitive because, by definition, an event occurs when the outcome of the random experiment is part of that event. However, since the empty set contains no elements, no outcome of the experiment can ever belong to it, making its probability zero.

The Difference Rule can be obtained by showing that $P(A) = P(A \cap B) + P(A - B)$. Note that the two sets $A \cap B$ and $A - B$ are disjoint and their union is A. Thus, by the third axiom of probability

$$\begin{aligned} P(A) &= P\left((A \cap B) \cup (A - B)\right) && \text{(since } A = (A \cap B) \cup (A - B)\text{)} \\ &= P(A \cap B) + P(A - B) && \text{(since } A \cap B \text{ and } A - B \text{ are disjoint)} \end{aligned}$$

The Addition Rule we obtain by noting that A and $B - A$ are disjoint sets and their union is $A \cup B$. Thus,

$$\begin{aligned} P(A \cup B) &= P(A \cup (B - A)) && \text{(since } A \cup B = A \cup (B - A)\text{)} \\ &= P(A) + P(B - A) && \text{(since } A \text{ and } B - A \text{ are disjoint)} \\ &= P(A) + P(B) - P(A \cap B) && \text{(by part the Difference Rule)} \end{aligned}$$

And finally the Subset Rule is a direct consequence of the fact that if $A \subset B$, then B can be written as the union of A and $B - A$. Since A and $B - A$ are disjoint, we have $P(B) = P(A) + P(B - A) \geq P(A)$.

We conclude this section with a few examples illustrating the application of these rules to calculate probabilities of joint events.

Example 2.6 A company has bid on two large construction projects. The company president believes that the probability of winning the first contract is 0.6, the probability of winning the second contract is 0.4, and the probability of winning both contracts is 0.2.

(a) What is the probability that the company wins at least one contract?
(b) What is the probability that the company wins the first contract but not the second contract?
(c) What is the probability that the company wins neither contract?
(d) What is the probability that the company wins exactly one contract?

Solution: Let A be the event that the company wins the first contract, and B be the event that the company wins the second contract. Given:

- $P(A) = 0.6$
- $P(B) = 0.4$
- $P(A \cap B) = 0.2$

(a) The probability that the company wins at least one contract is the probability of the union of events A and B. Using the Addition Rule:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.6 + 0.4 - 0.2 \\ &= 0.8 \end{aligned}$$

Therefore, the probability that the company wins at least one contract is 0.8.

(b) The probability that the company wins the first contract but not the second contract is the probability of the difference between events A and B. Using the Difference Rule:

$$\begin{aligned} P(A - B) &= P(A) - P(A \cap B) \\ &= 0.6 - 0.2 \\ &= 0.4 \end{aligned}$$

Therefore, the probability that the company wins the first contract but not the second contract is 0.4.

(c) The probability that the company wins neither contract is the probability of the complement of the union

of events A and B. Using the Complement Rule:

$$\begin{aligned} P((A \cup B)^c) &= 1 - P(A \cup B) \\ &= 1 - 0.8 \\ &= 0.2 \end{aligned}$$

Therefore, the probability that the company wins neither contract is 0.2.

(d) The probability that the company wins exactly one contract is the probability of the difference between the union of events A and B and the intersection of events A and B. Using the Difference Rule:

$$\begin{aligned} P((A \cup B) - (A \cap B)) &= P(A \cup B) - P(A \cap B) \\ &= 0.8 - 0.2 \\ &= 0.6 \end{aligned}$$

So, the probability that the company wins exactly one contract is 0.6.

◀

Example 2.7

- There is a 60 percent chance that it will rain today.
- There is a 50 percent chance that it will rain tomorrow.
- There is a 30 percent chance that it does not rain either day.

(a) The probability that it will rain today or tomorrow

(b) The probability that it will rain today and tomorrow.

(c) The probability that it will rain today but not tomorrow.

(d) The probability that it either will rain today or tomorrow, but not both.

Solution:

(a) Let A be the event that it rains today, and B be the event that it rains tomorrow.
Given:

$$\begin{aligned} P(A \cup B) &= 1 - P\left((A \cup B)^c\right) && \text{by the Complement Rule} \\ &= 1 - P\left(A^c \cap B^c\right) && \text{by De Morgan's Law} \\ &= 1 - 0.3 \\ &= 0.7 \end{aligned}$$

Therefore, the probability that it will rain today or tomorrow is 0.7.

(b) The probability that it will rain today and tomorrow: this is $P(A \cap B)$. To find this we note that

$$\begin{aligned} P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\ &= 0.6 + 0.5 - 0.7 \\ &= 0.4 \end{aligned}$$

(c) The probability that it will rain today but not tomorrow: this is $P\left(A \cap B^c\right)$.

$$\begin{aligned} P(A \cap B^c) &= P(A - B) \\ &= P(A) - P(A \cap B) \\ &= 0.6 - 0.4 \\ &= 0.2 \end{aligned}$$

(d) The probability that it either will rain today or tomorrow but not both: this is $P(A - B) + P(B - A)$. We have already found $P(A - B) = .2$. Similarly, we can find $P(B - A)$:

$$\begin{aligned} P(B - A) &= P(B) - P(B \cap A) \\ &= 0.5 - 0.4 \\ &= 0.1 \end{aligned}$$

Thus,

$$\begin{aligned} P(A - B) + P(B - A) &= 0.2 + 0.1 \\ &= 0.3 \end{aligned}$$

◀

2.4 Conditional Probability

Consider the following scenario: in a particular population, 5% of individuals have a specific medical condition. Therefore, the probability that a randomly selected person has the condition is 5%:

$$P(D) = 0.05,$$

where D represents the event that a person has the disease. This probability $P(D)$ is known as the *prior probability*, as it reflects our initial belief about the likelihood of the event before any additional information is obtained.

Now, imagine selecting a random person and being informed that they have tested positive for the disease. With this additional information, how should we update the probability that the person actually has the disease? In other words, what is the probability that a person has the disease given that they tested positive? Let T denote the event that a person tests positive. This conditional probability is expressed as:

$$P(D \mid T),$$

which represents the probability of D occurring given that T has occurred. This updated probability $P(D \mid T)$ is known as the *posterior probability*, as it reflects our revised belief about the likelihood of the event after taking the new evidence into account. Intuitively, it is reasonable to expect that $P(D \mid T)$ is greater than the prior probability $P(D)$. However, what is the exact value of $P(D \mid T)$? Before introducing a general formula, let us consider a simple example.

Example 2.8 I roll a fair die. Let A be the event that the outcome is a prime number, i.e., $A = \{2, 3, 5\}$. Also, let B be the event that the outcome is greater than or equal to 3, i.e., $B = \{3, 4, 5\}$.

(a) What is the probability of A, $P(A)$?

(b) What is the probability of A given B, $P(A \mid B)$?

Solution:

(a) The probability of A is the number of outcomes in A divided by the total number of outcomes. Since there are 3 prime numbers and 6 possible outcomes, we have:

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

(b) The probability of A given B is the number of outcomes in the intersection of A and B divided by the number of outcomes in B. Since the outcomes in the intersection of A and B are $\{3, 5\}$ and the outcomes in B are $\{3, 4, 5\}$, we have:

$$P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{|\{3, 5\}|}{|\{3, 4, 5\}|} = \frac{2}{3}$$

◀

Having understood the basic example, we can now generalize the approach to derive a more universal formula for conditional probability. Starting from the specific case, we can manipulate the expression by dividing both the numerator and the denominator by the total number of possible outcomes, $|S|$, as shown below:

$$P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{\frac{|A \cap B|}{|S|}}{\frac{|B|}{|S|}} = \frac{P(A \cap B)}{P(B)}$$

Explanation:

- **Numerator:** $|A \cap B|$ represents the number of outcomes where both events A and B occur. Dividing by $|S|$ converts this count into a probability, $P(A \cap B)$.
- **Denominator:** $|B|$ is the number of outcomes where event B occurs. Similarly, dividing by $|S|$ yields the probability of event B, denoted as $P(B)$.

Thus, the conditional probability $P(A \mid B)$ can be expressed as the ratio of the joint probability of A and B to the probability of B.

While the above derivation assumes a finite sample space with equally likely outcomes, the resulting formula is remarkably general. It holds true regardless of whether the sample space is finite or infinite, and whether the outcomes are equally likely or not. This universality makes the formula a cornerstone of probability theory.

Definition 2.4 (Conditional Probability)

For any two events A and B with $P(B) > 0$, the conditional probability of A given B is defined as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0$$

♣

Here is the intuition behind the formula. When we know that event B has occurred, we effectively eliminate all outcomes that are not part of B. This reduction transforms our original sample space into the subset B.

Within this new, restricted sample space B, the only way for event A to occur is if the outcome lies in the intersection of A and B, denoted by $A \cap B$. To determine the conditional probability $P(A \mid B)$, we calculate

the ratio of the probability of both A and B occurring to the probability of B occurring alone. Mathematically, this is expressed as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

This formulation ensures that the probabilities within the new sample space B are normalized, meaning the total probability sums to 1. For instance, consider the conditional probability of B given B:

$$P(B \mid B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

This result intuitively confirms that if B has occurred, the probability of B occurring is certain, i.e., 1.

Note that the elementary formula for $P(A \mid B)$ is undefined when $P(B) = 0$, because it would require division by zero. A probability-zero event need not be empty in a continuous probability model, but conditioning on such events requires more advanced definitions beyond the scope of this chapter.

We can also formulate the axioms of probability in terms of conditional probability. The axioms of probability are as follows:

Axiom 2.2 (Axioms of Conditional Probability)

- **Axiom 1:** For any event A, $0 \leq P(A \mid B) \leq 1$.
- **Axiom 2:** Conditional probability of B given B is 1, i.e., $P(B \mid B) = 1$.
- **Axiom 3:** If $A_1, A_2, A_3, \cdots$ are disjoint events, then
$$P\left(A_1 \cup A_2 \cup A_3 \cdots \mid B\right) = P\left(A_1 \mid B\right) + P\left(A_2 \mid B\right) + P\left(A_3 \mid B\right) + \cdots$$

♡

And we are also able to derive other rules of conditional probability from these axioms:

Theorem 2.2 (Rules of Conditional Probability)

- **Complement Rule:** The probability of the complement of event A given C is
$$P\left(A^c \mid C\right) = 1 - P(A \mid C)$$
- **Empty Set Rule:** The probability of the empty set given some event C is 0, i.e.,
$$P(\emptyset \mid C) = 0$$
- **Addition Rule:** For any two events A and B, the probability of the union of events A and B given another event C is
$$P(A \cup B \mid C) = P(A \mid C) + P(B \mid C) - P(A \cap B \mid C)$$
- **Difference Rule:** The probability of the difference between events A and B given another event C is
$$P(A - B \mid C) = P(A \mid C) - P(A \cap B \mid C)$$
- **Subset Rule:** If $A \subset B$, then
$$P(A \mid C) \leq P(B \mid C)$$

♡

The following example illustrates the application of conditional probability and introduces the concept of **contingency tables** as well as **false positives** and **false negatives**.

Example 2.9 A researcher aims to assess the effectiveness of a diagnostic test designed to detect renal disease in patients with high blood pressure. To achieve this, she conducts the test on a sample of 137 patients, categorized as follows:

- **67 patients** with a confirmed diagnosis of renal disease.
- **70 patients** who are known to be healthy (i.e., do not have renal disease).

The diagnostic test produces one of two possible outcomes for each patient:

- **Positive**: Indicates that the patient has renal disease.
- **Negative**: Indicates that the patient does not have renal disease.

The findings are summarised in the following contingency table:

	Test Results		
Truth	Positive	Negative	Total
Renal Disease	44	23	67
Healthy	10	60	70
Total	54	83	137

In this experiment:

- **True Positives**: Patients who have renal disease and tested positive.
- **False Negatives**: Patients who have renal disease but tested negative.
- **False Positives**: Healthy patients who tested positive.
- **True Negatives**: Healthy patients who tested negative.

Determine the following probabilities:

(a) The Probability of having renal disease, $P(D)$.
(b) The Probability of a positive test, $P(T^+)$.
(c) The Probability of a negative test, $P(T^-)$.
(d) If a person has renal disease, what is the probability that they test positive for the disease?
(e) Determine the probability that a patient has renal disease given a positive test result, $P(D \mid T^+)$.
(f) Determine the probability that a patient does not have renal disease given a negative test result, $P(\text{Healthy} \mid T^-)$.
(g) Assess the overall accuracy of the diagnostic test.

Solution: Using the contingency table, we can calculate the following probabilities:

(a) **Probability of Renal Disease,** $P(D)$:

$$P(D) = \frac{\text{Number of patients with renal disease}}{\text{Total number of patients}}$$
$$= \frac{67}{137} \approx 0.4883 \text{ or } 48.83\%$$

(b) **Probability of a Positive Test,** $P(T^+)$:

$$P(T^+) = \frac{\text{Number of positive tests}}{\text{Total number of patients}}$$
$$= \frac{54}{137} \approx 0.3942 \text{ or } 39.42\%$$

(c) **Probability of a Negative Test,** $P(T^-)$:

$$P(T^-) = \frac{\text{Number of negative tests}}{\text{Total number of patients}}$$
$$= \frac{83}{137} \approx 0.6058 \text{ or } 60.58\%$$

(d) **Probability of a Positive Test Given Renal Disease,** $P(T^+ \mid D)$:

$$P(T^+ \mid D) = \frac{44}{67} \approx 0.6567$$

(e) **Probability of Renal Disease Given a Positive Test,** $P(D \mid T^+)$:

$$P(D \mid T^+) = \frac{P(D \cap T^+)}{P(T^+)} = \frac{44}{54} = \frac{22}{27} \approx 0.8148$$

(f) **Probability of Being Healthy Given a Negative Test,** $P(\textbf{Healthy} \mid T^-)$:

$$P(\text{Healthy} \mid T^-) = \frac{P(\text{Healthy} \cap T^-)}{P(T^-)} = \frac{60}{83} \approx 0.7229$$

(g) **Overall Accuracy of the Diagnostic Test**: The overall accuracy of the diagnostic test is the proportion of correct diagnoses, i.e., the sum of true positives and true negatives divided by the total number of patients:

$$\text{Overall Accuracy} = \frac{44 + 60}{137} = \frac{104}{137} \approx 0.7591$$

The calculations describe several different performance measures. The sensitivity, which is the probability of a positive result among patients with renal disease, is $44/67 \approx 65.67\%$. The specificity, which is the probability of a negative result among healthy patients, is $60/70 \approx 85.71\%$. By contrast, $P(D \mid T^+) = 81.48\%$ is the positive predictive value, and $P(\text{Healthy} \mid T^-) = 72.29\%$ is the negative predictive value. The overall accuracy is 75.91%. ◀

Here's a classic probability puzzle known as the Two-Child Problem. This scenario has appeared in various forms in the literature, each with subtle twists that lead to different results. Before looking at the calculations, take a moment to make your own predictions — you might be surprised by the outcomes!

Example 2.10 Consider a family with two children, and we are interested in the possible biological gender combinations of the children. The sample space for this situation is:

$$S = \{(G, G), (G, B), (B, G), (B, B)\}$$

where G represents a biological girl (hence 'girl') and B represents a biological boy (hence 'boy'). For simplicity, we assume that all four outcomes are equally likely.

(a) What is the probability that both children are girls given that the first child is a girl?

(b) We ask the father: "Do you have at least one daughter?" He responds "Yes!" Given this extra information, what is the probability that both children are girls? In other words, what is the probability that both children are girls given that we know at least one of them is a girl?

Solution: Let A be the event that both children are girls, i.e., $A = \{(G, G)\}$. Let B be the event that the first child is a girl, i.e., $B = \{(G, G), (G, B)\}$. Finally, let C be the event that at least one of the children is a girl,

i.e., $C = \{(G,G),(G,B),(B,G)\}$. Since the outcomes are equally likely, we can write

$$P(A) = \frac{1}{4}$$
$$P(B) = \frac{2}{4} = \frac{1}{2}$$
$$P(C) = \frac{3}{4}$$

(a) What is the probability that both children are girls given that the first child is a girl? This is $P(A \mid B)$, thus we can write

$$\begin{aligned} P(A \mid B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(A)}{P(B)} \quad (\text{ since } A \subset B) \\ &= \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2} \end{aligned}$$

(b) What is the probability that both children are girls given that we know at least one of them is a girl? This is $P(A \mid C)$, thus we can write

$$\begin{aligned} P(A \mid C) &= \frac{P(A \cap C)}{P(C)} \\ &= \frac{P(A)}{P(C)} \quad (\text{ since } A \subset C) \\ &= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3} \end{aligned}$$

◀

Remark: Many people might intuitively guess that both $P(A \mid B)$ and $P(A \mid C)$ would be 50%. However, while $P(A \mid B) = 50\%$, $P(A \mid C)$ is only 33%. This illustrates how probability can be counterintuitive. The key is to recognize that event B is a subset of event C. Specifically, B excludes the outcome (B,G), which is included in C. As a result, C has more outcomes not in A than B, leading to a smaller $P(A \mid C)$ compared to $P(A \mid B)$.

The result in part (b) assumes that the information supplied is exactly the event C, "at least one child is a girl." If the father was selected or questioned using a different random procedure, the conditional probability can change. The information-generating procedure must therefore be specified when interpreting this puzzle.

2.5 Multiplication and Total Probability Rules

The multiplication rule is a fundamental principle in probability theory that allows us to calculate the probability of multiple events occurring in sequence. This rule is particularly useful when the events are dependent, meaning that the occurrence of one event influences the probability of the subsequent event.

Theorem 2.3 (Multiplication Rule)

If $P(A) > 0$ and $P(B) > 0$, the probability of both events occurring is given by:

$$P(A \cap B) = P(A)P(B \mid A) = P(B)P(A \mid B)$$

♡

Example 2.11 Suppose a software development team is testing a new feature. There are two stages of testing: Unit Testing (A) and Integration Testing (B). The probability that a bug is detected in Unit Testing is $P(A) = 0.3$. If a bug is detected during Unit Testing, the probability that it will also be detected in Integration Testing is $P(B \mid A) = 0.7$. What is the probability that a bug is detected in both stages of testing?

Solution: Using the Multiplication Rule:

$$P(A \cap B) = P(A) \cdot P(B \mid A) = 0.3 \times 0.7 = 0.21$$

Therefore, the probability that a bug is detected in both Unit Testing and Integration Testing is 0.21. ◀

Example 2.12 A company has implemented a two-layer security system to protect against network breaches: **Firewall (A)** and **Intrusion Detection System (IDS) (B)**. The probability that a breach attempt is detected by the Firewall is $P(A) = 0.4$. If the breach passes through the Firewall, the probability that it is detected by the IDS is $P(B \mid A^c) = 0.6$. Given that the Firewall detects the breach, the conditional probability that the IDS also detects it is $P(B \mid A) = 0.8$.

(a) What is the probability that a breach is detected by at least one of the security layers?
(b) What is the probability that a breach is detected by both security layers?

Solution:

(a) **Probability of Detection by at Least One Layer:**
To find the probability that the breach is detected by at least one layer, we calculate the probability of a breach passing through both layers undetected and subtract it from 1.

$$P(\text{Undetected}) = P(A^c) \cdot P(B^c \mid A^c) = (1 - 0.4) \cdot (1 - 0.6) = 0.6 \cdot 0.4 = 0.24$$

Therefore, the probability of detecting the breach with at least one layer is:

$$P(\text{Detected}) = 1 - P(\text{Undetected}) = 1 - 0.24 = 0.76$$

(b) **Probability of Detection by Both Layers:**
Using the Multiplication Rule:

$$P(A \cap B) = P(A) \cdot P(B \mid A) = 0.4 \cdot 0.8 = 0.32$$

The probability of detecting the breach with at least one layer is 0.76, while the probability of detecting the breach with by both the Firewall and the IDS is 0.32. This discrepancy highlights the importance of considering the dependencies between events when calculating probabilities.

◀

Remark: It might seem intuitive to some to calculate the joint probability of detection using $P(A \cap B) = P(A) \cdot P(B) = 0.4 \cdot 0.6 = 0.24$. However, this approach ignores the fact that the probability of detection by the IDS depends on the outcome of the Firewall. This is why we use the conditional probability $P(B \mid A)$, which properly accounts for the dependency between the two layers, yielding the correct result of 0.32.

In some scenarios, the probability of an event depends on various conditions. By knowing the conditional probabilities under these different scenarios, we can determine the overall probability of the event. For instance,

consider semiconductor manufacturing, where we define:

- A as the event that a chip is highly contaminated
- B as the event that a product using the chip fails

The probability of failure for a non-contaminated chip is $P(B \mid A^c) = 0.005$. On the other hand, if the chip is subjected to high levels of contamination, the probability of failure is $P(B \mid A) = 0.10$. Suppose that in a given production run, 20% of the chips are highly contaminated ($P(A) = 0.20$). What is the probability that a product using one of these chips fails?

The probability of failure depends on whether the chip was exposed to high contamination or not. For any event A, it can be decomposed into two mutually exclusive parts: one that intersects with B and another that intersects with the complement B^c. Mathematically, we can express this as:

$$A = (B \cap A) \cup (B^c \cap A)$$

This decomposition is visualized in the Venn diagram in Figure 2.1. Since B and B^c are mutually exclusive, their intersections with A are also mutually exclusive. Thus, using the rule for the probability of the union of mutually exclusive events and the multiplication rule, we can derive the total probability as follows:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

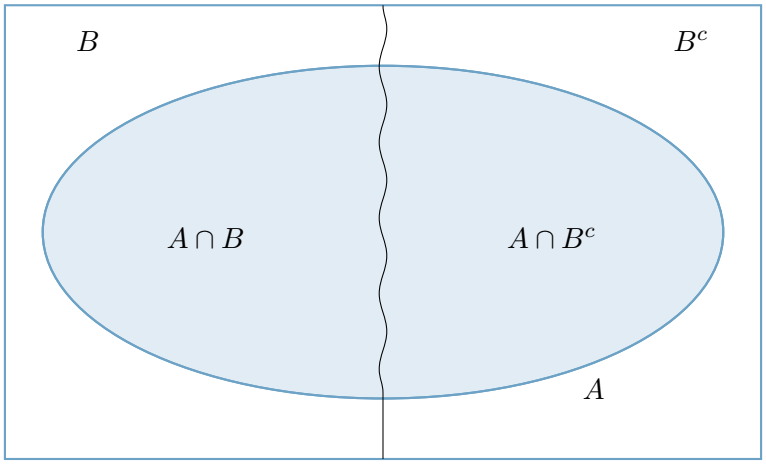


Figure 2.1: $P(A) = P(A \cap B) + P(A \cap B^c)$

We summarise this in the following theorem:

Theorem 2.4 (Law of Total Probability)

For any events A and B such that B and B^c form a partition of the sample space S, the total probability of event A is given by:

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B^c) \\ &= P(B)P(A \mid B) + P(B^c)P(A \mid B^c) \end{aligned}$$

For any event A and any partition of the sample space $B_1, B_2, \ldots, B_n$ such that $B_i \cap B_j = \emptyset$ for all $i \neq j$ and $\bigcup_{i=1}^{n} B_i = S$, the total probability of event A is given by:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i)$$



Let us consider some examples.

Example 2.13 A company produces two types of products: **Product A** and **Product B**. The probability that a product is defective is $P(D \mid A) = 0.05$ for Product A and $P(D \mid B) = 0.10$ for Product B. The company manufactures 60% of its products as Product A and 40% as Product B. What is the probability that a randomly selected product is defective?

Solution: Using the Law of Total Probability:

$$P(D) = P(D \mid A)P(A) + P(D \mid B)P(B)$$

Substituting the given values:

$$P(D) = 0.05 \times 0.60 + 0.10 \times 0.40 = 0.03 + 0.04 = 0.07$$

Therefore, the probability that a randomly selected product is defective is 0.07. ◀

A graphical display of partitioning an event B among a collection of mutually exclusive and exhaustive events is shown in Figure 2.2. The event A is partitioned into five mutually exclusive events B_1, B_2, B_3, B_4, B_5, which together form the sample space S. The total probability of event A is calculated as the sum of the conditional probabilities of A given each partition B_i multiplied by the probability of each partition $P(B_i)$.

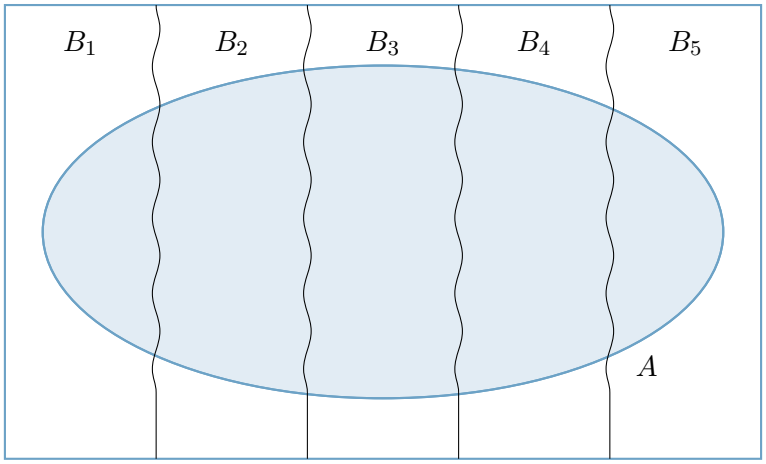


Figure 2.2: $P(A) = \sum_{i=1}^{5} P(A \mid B_i)P(B_i)$

Example 2.14 A university offers three types of courses: **Online**, **Hybrid**, and **In-Person**. The probability that a student fails a course is $P(F \mid \text{Online}) = 0.10$ for Online courses, $P(F \mid \text{Hybrid}) = 0.05$ for Hybrid courses, and $P(F \mid \text{In-Person}) = 0.02$ for In-Person courses. The university offers 50% of its courses as Online, 30% as Hybrid, and 20% as In-Person. What is the probability that a randomly selected student fails a course?

Solution: Using the Law of Total Probability:

$$P(F) = P(F \mid \text{Online})P(\text{Online}) + P(F \mid \text{Hybrid})P(\text{Hybrid}) + P(F \mid \text{In-Person})P(\text{In-Person})$$

Substituting the given values:

$$P(F) = 0.10 \times 0.50 + 0.05 \times 0.30 + 0.02 \times 0.20 = 0.05 + 0.015 + 0.004 = 0.069$$

Therefore, the probability that a randomly selected student fails a course is 0.069. ◀

2.6 Independence

Let A represent the event that it rains tomorrow, with $P(A) = \frac{1}{3}$. Additionally, suppose I toss a fair coin, and let B be the event that it lands heads up, so $P(B) = \frac{1}{2}$.

Now, consider the probability $P(A \mid B)$. What do you think it would be? You might intuitively guess that $P(A \mid B) = P(A) = \frac{1}{3}$, and you would be correct! The coin toss outcome has no influence on the weather forecast. This means that whether B occurs or not, the probability of A remains unchanged. This scenario illustrates the concept of independent events: two events are independent if the occurrence of one does not provide any information about the other.

Let's now formalize the definition of independence.

Definition 2.5 (Independence)

Two events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

When $P(B) > 0$, this is equivalent to $P(A \mid B) = P(A)$, and when $P(A) > 0$, it is equivalent to $P(B \mid A) = P(B)$.

♣

In summary, independence can be understood in two equivalent ways: "Independence means that the probability of the intersection of two events can be found by simply multiplying their individual probabilities," or, alternatively, "Independence means that the conditional probability of one event given the other is the same as the original, unconditioned probability."

Lemma 2.1

If A and B are independent then

- A and B^c are independent,
- A^c and B are independent,
- A^c and B^c are independent.

♡

When dealing with the probability of the union of multiple independent events, $A_1, A_2, \ldots, A_n$, it is often easier to find the probability of their intersection than their union. In these situations, De Morgan's Law can be quite useful:

$$A_1 \cup A_2 \cup \cdots \cup A_n = (A_1^c \cap A_2^c \cap \cdots \cap A_n^c)^c$$

Using this relationship, we can express the probability of the union as:

$$\begin{aligned} P(A_1 \cup A_2 \cup \cdots \cup A_n) &= 1 - P(A_1^c \cap A_2^c \cap \cdots \cap A_n^c) \\ &= 1 - P(A_1^c) P(A_2^c) \cdots P(A_n^c) \quad \text{(since the } A_i\text{'s are independent)} \\ &= 1 - (1 - P(A_1))(1 - P(A_2)) \cdots (1 - P(A_n)). \end{aligned}$$

Theorem 2.5 (Independence and DeMorgan's Law)

If $A_1, A_2, \cdots, A_n$ are independent then

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = 1 - (1 - P(A_1))(1 - P(A_2)) \cdots (1 - P(A_n))$$

♡

Warning! A common misconception is to confuse independence with disjointness. However, these are fundamentally different concepts. Two events, A and B, are disjoint if $A \cap B = \emptyset$. If both events have positive probability, disjointness implies dependence: knowing that one occurred rules out the other. The exception is when at least one of the disjoint events has probability zero; in that case, the product criterion for independence is satisfied.

Concept	Description	Key Formulas
Disjoint	Events A and B cannot occur at the same time	$A \cap B = \emptyset$ $P(A \cup B) = P(A) + P(B)$
Independent	Occurrence of B gives no information about A	$P(A \mid B) = P(A)$ $P(B \mid A) = P(B)$ $P(A \cap B) = P(A) \cdot P(B)$

Table 2.2: Comparison of Disjoint and Independent Events.

We can extend the concept of independence to multiple events. A set of events $A_1, A_2, \ldots, A_n$ are considered independent if the occurrence of any subset of these events does not provide any information about the occurrence of the other events. Mathematically, this is expressed as:

Definition 2.6 (Independence of Multiple Events)

Events $A_1, A_2, \ldots, A_k$ are mutually independent if, for every subset $J \subseteq \{1, \ldots, k\}$ containing at least two indices,

$$P\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} P(A_j).$$

Requiring the equality only for the intersection of all k events is not sufficient.

Example 2.15 The Gambler's Fallacy

The **Gambler's Fallacy** is a common cognitive bias that arises when individuals believe that the outcome of a random event is influenced by previous outcomes. This fallacy is often observed in gambling scenarios, where individuals incorrectly assume that the probability of an event occurring is affected by past events.

A man tosses a fair coin eight times and observes whether the toss yields a head (H) or a tail (T) on each toss. Which of the following sequences of coin tosses is the man more likely to get a head (H) on his next toss? This one:

TTTTTTTT

or this one:

HHTHTTHH

The answer is neither as illustrated here:

$$P\left(H_9 \mid T_1T_2\ldots T_8\right) = \frac{P\left(H_9 \cap T_1 \cap T_2 \cap \cdots \cap T_8\right)}{P\left(T_1 \cap T_2 \cap \cdots \cap T_8\right)} = \frac{\left(\frac{1}{2}\right)^9}{\left(\frac{1}{2}\right)^8} = \frac{1}{2}$$

Try to avoid falling into the trap of the Gambler's Fallacy. For example, if a fair coin lands on tails eight times in a row, it's easy to think that a head is "due" on the next toss. However, each toss is independent, and the probability remains the same — there's still a 50% chance of heads or tails, regardless of past results.

2.7 Bayes' Theorem

We are now ready to introduce one of the most powerful tools in conditional probability: Bayes' rule (or theorem). This rule is particularly useful when we know $P(A \mid B)$ but want to find $P(B \mid A)$. Starting with the definition of conditional probability, we have:

$$P(A \mid B) \cdot P(B) = P(A \cap B) = P(B \mid A) \cdot P(A)$$

By dividing both sides by $P(A)$, we arrive at:

$$P(B \mid A) = \frac{P(A \mid B) \cdot P(B)}{P(A)}$$

This formula is famously known as Bayes' rule. In many cases, to calculate $P(A)$ in Bayes' rule, we need the law of total probability. Therefore, Bayes' rule is often presented in the form:

$$P(B \mid A) = \frac{P(A \mid B) \cdot P(B)}{P(A \mid B) \cdot P(B) + P(A \mid B^c) \cdot P(B^c)}$$

or more generally

$$P\left(B_j \mid A\right) = \frac{P\left(A \mid B_j\right) \cdot P\left(B_j\right)}{\sum_i P\left(A \mid B_i\right) \cdot P\left(B_i\right)}$$

where $B_1, B_2, \ldots, B_n$ form a partition of the sample space.

Theorem 2.6 (Bayes' Theorem)

For any two events A and B, where $P(A) \neq 0$, we have

$$P(B \mid A) = \frac{P(A \mid B) \cdot P(B)}{P(A)} = \frac{P(A \mid B) \cdot P(B)}{P(A \mid B) \cdot P(B) + P(A \mid B^c) \cdot P(B^c)}$$

If $B_1, B_2, B_3, \cdots$ form a partition of the sample space S, and A is any event with $P(A) \neq 0$, we have

$$P\left(B_j \mid A\right) = \frac{P\left(A \mid B_j\right) \cdot P\left(B_j\right)}{\sum_i P\left(A \mid B_i\right) \cdot P\left(B_i\right)}$$

♡

We start with an infamous example to highlight the importance of Bayes' Theorem.

Example 2.16 False Positive Paradox

Imagine a rare disease that affects about 1 in every 10,000 people. There is a test available to detect this disease, and while the test is highly accurate, it is not perfect. Specifically:

- The probability that the test shows a positive result (indicating the disease) when the person does not have the disease is 2%.
- The probability that the test shows a negative result (indicating no disease) when the person does have the disease is 1%.

Now, suppose a randomly selected person takes the test, and the result comes back positive. What is the probability that this person actually has the disease?

Solution: Let D be the event that the person has the disease, let T^+ be the event that the test result is positive, and let T^- be the event that it is negative. We are given:

$$P(D) = \frac{1}{10,000}$$
$$P\left(T^+ \mid D^c\right) = 0.02 \quad \text{(False Positive Rate)}$$
$$P\left(T^- \mid D\right) = 0.01 \quad \text{(False Negative Rate)}$$

We want to compute $P(D \mid T^+)$. First, we need $P(T^+ \mid D)$, which is the probability of a true positive. Since a person with the disease will either test positive or negative, we have:

$$P(T^+ \mid D) = 1 - P(T^- \mid D) = 1 - 0.01 = 0.99$$

Now, using Bayes' rule:

$$\begin{aligned} P(D \mid T^+) &= \frac{P(T^+ \mid D)P(D)}{P(T^+ \mid D)P(D) + P\left(T^+ \mid D^c\right)P\left(D^c\right)} \\ &= \frac{0.99 \times 0.0001}{0.99 \times 0.0001 + 0.02 \times (1 - 0.0001)} \\ &= \frac{0.000099}{0.000099 + 0.02 \times 0.9999} \\ &= \frac{0.000099}{0.000099 + 0.019998} \\ &\approx 0.0049 \end{aligned}$$

This result means there is less than half a percent chance that the person actually has the disease. Despite the positive test result, the low prevalence of the disease and the test's false positive rate contribute to this counterintuitive outcome. ◀

Example 2.17 Bayesian networks are commonly used on the websites of high-technology manufacturers to help customers quickly diagnose issues with their products. For instance, a printer manufacturer uses data from test results to identify potential causes of printer failures. Printer failures are primarily associated with three types of problems: hardware, software, and other issues (like connectors). The probabilities of these problems are as follows:

- Probability of a hardware issue: $P(H) = 0.1$
- Probability of a software issue: $P(S) = 0.6$
- Probability of another type of issue: $P(O) = 0.3$

The likelihood of a printer failing, given each type of problem, is:

- Probability of failure given a hardware issue: $P(F \mid H) = 0.9$
- Probability of failure given a software issue: $P(F \mid S) = 0.2$

- Probability of failure given another issue: $P(F \mid O) = 0.5$

Given that a customer experiences a printer failure and uses the manufacturer's website to diagnose the issue, what is the most likely cause of the problem?

Solution: To determine the most likely cause of the printer failure, we need to calculate the probability of each type of problem given that a failure has occurred. This involves using Bayes' theorem to compute the posterior probabilities.

Let F denote the event of a printer failure. We want to find:

$$P(H \mid F), \quad P(S \mid F), \quad P(O \mid F)$$

Step 1: Calculate the Total Probability of Failure

First, we use the law of total probability to find $P(F)$:

$$P(F) = P(F \mid H) \cdot P(H) + P(F \mid S) \cdot P(S) + P(F \mid O) \cdot P(O)$$

Substituting the given values:

$$P(F) = (0.9 \times 0.1) + (0.2 \times 0.6) + (0.5 \times 0.3)$$
$$P(F) = 0.09 + 0.12 + 0.15 = 0.36$$

Step 2: Apply Bayes' Theorem to Find the Posterior Probabilities

Now, apply Bayes' theorem for each problem type:

1. Probability of Hardware Problem Given Failure:

$$P(H \mid F) = \frac{P(F \mid H) \cdot P(H)}{P(F)} = \frac{0.9 \times 0.1}{0.36} = \frac{0.09}{0.36} = 0.25$$

2. Probability of Software Problem Given Failure:

$$P(S \mid F) = \frac{P(F \mid S) \cdot P(S)}{P(F)} = \frac{0.2 \times 0.6}{0.36} = \frac{0.12}{0.36} = 0.3333$$

3. Probability of Other Problems Given Failure:

$$P(O \mid F) = \frac{P(F \mid O) \cdot P(O)}{P(F)} = \frac{0.5 \times 0.3}{0.36} = \frac{0.15}{0.36} = 0.4167$$

Step 3: Interpret the Results

- Hardware Problem: 25% chance
- Software Problem: Approximately 33.33% chance
- Other Problem: Approximately 41.67% chance

Conclusion: Given a printer failure, the most likely cause is an Other Problem, such as connectors, with a probability of approximately 41.67%. ◀

Chapter 3 Random Variables and Discrete Distributions

In chapter 2, probabilities were assigned to events: a component fails, a test returns a positive result, or a system experiences a particular combination of outcomes. In many applications, however, the outcome of interest is numerical. We may want to record the number of failed requests in an hour, the number of defective components in a batch, or the number of attempts required before a transmission succeeds. To analyse such quantities systematically, we introduce the concept of a **random variable**.

A random variable connects the outcomes of a random experiment with numbers. This allows an uncertain process to be described using familiar mathematical ideas such as functions, sums, averages, and variation. The possible values of the variable do not all need to be equally likely. Their probabilities form a distribution, which provides a complete probabilistic description of the variable and makes it possible to answer questions about both individual outcomes and long-run behaviour.

This chapter focuses on **discrete random variables**, whose possible values form a finite or countable set. A packet is either lost or delivered; a test suite contains an integer number of failures; and a queue receives a countable number of requests during a specified interval. These examples lead naturally to three important models. The Bernoulli distribution describes a single trial with two possible outcomes. The binomial distribution describes the number of successes in a fixed number of independent Bernoulli trials. The Poisson distribution describes counts occurring within a fixed interval of time, space, or another exposure measure.

Before studying these models, we develop the general language used to describe discrete random variables. Probability mass functions assign probabilities to individual values, while cumulative distribution functions describe probabilities up to a specified threshold. Expected value and variance then summarise the centre and spread of a distribution. Together, these concepts provide the foundation for selecting, interpreting, and applying discrete probability models throughout statistics and data analysis.

3.1 Random variables, PMFs, and CDFs

An event describes a collection of outcomes, but it does not necessarily describe them numerically. For example, suppose that two components are inspected and that each component either passes or fails. The sample space may be written as

$$S = \{PP, PF, FP, FF\},$$

where PF means that the first component passes and the second fails. If the quantity of interest is the *number* of failed components, the four outcomes are represented by the numbers 0, 1, and 2. A random variable formalises this assignment.

Definition 3.1 (Random Variable)

Let S be the sample space of a random experiment. A **random variable** is a function

$$X : S \longrightarrow \mathbb{R}$$

that assigns a real number $X(\omega)$ to each outcome $\omega \in S$.

♣

The word *variable* can be slightly misleading. A random variable is a function with a fixed rule; it is the

outcome of the underlying experiment, and therefore the resulting numerical value, that is uncertain. Before the experiment is observed, the random variable is denoted by an uppercase letter such as X. A particular realised value is denoted by the corresponding lowercase letter x.

Example 3.1 Number of Failed Components

Let X denote the number of failed components when two components are inspected. The mapping from the sample space to the real numbers is

$$X(PP) = 0, \qquad X(PF) = 1, \qquad X(FP) = 1, \qquad X(FF) = 2.$$

Notice that different outcomes can produce the same value of X. In particular, the event that exactly one component fails is

$$\{X = 1\} = \{PF, FP\}.$$

Thus, a statement involving a random variable defines an event in the original sample space.

Random variables may be discrete or continuous. A discrete random variable takes values in a finite or countably infinite set. Counts are therefore discrete, even when there is no fixed upper bound. A continuous random variable can take values throughout an interval and will be studied in chapter 4.

Definition 3.2 (Discrete Random Variable and Support)

A random variable X is **discrete** if its possible values form a finite or countably infinite set. The set of values with positive probability is called the **support** of X:

$$\operatorname{supp}(X) = \{x \in \mathbb{R} : P(X = x) > 0\}.$$

♣

For the component example, $\operatorname{supp}(X) = \{0, 1, 2\}$. The support records the possible numerical values, while the probability distribution specifies how probability is allocated among them.

The probability mass function

The distribution of a discrete random variable is described by its probability mass function. It gives the probability of every individual value in the support.

Definition 3.3 (Probability Mass Function)

The **probability mass function** (PMF) of a discrete random variable X is the function

$$p_X(x) = P(X = x).$$

For values outside the support of X, $p_X(x) = 0$.

♣

Theorem 3.1 (Properties of a Probability Mass Function)

A function p_X is a valid PMF if and only if

1. $p_X(x) \geq 0$ for every x, and
2. the probabilities over all values in the support sum to one:
$$\sum_{x \in \text{supp}(X)} p_X(x) = 1.$$

For any set $A \subseteq \mathbb{R}$,
$$P(X \in A) = \sum_{x \in A \cap \text{supp}(X)} p_X(x).$$

♡

The final formula is particularly important: probabilities involving a discrete random variable are found by adding the probability masses of the values that satisfy the event.

Example 3.2 PMF for Failed Components

Suppose the two component inspections are independent and that each component fails with probability 0.1. The probabilities associated with the four outcomes are

$$P(PP) = 0.9 \cdot 0.9 = 0.81,$$
$$P(PF) = 0.9 \cdot 0.1 = 0.09,$$
$$P(FP) = 0.1 \cdot 0.9 = 0.09,$$
$$P(FF) = 0.1 \cdot 0.1 = 0.01.$$

Since X counts the number of failures, its PMF is

x	0	1	2
$p_X(x)$	0.81	0.18	0.01

where $p_X(1) = P(PF) + P(FP) = 0.18$. The values are non-negative and

$$0.81 + 0.18 + 0.01 = 1,$$

so the PMF is valid. The probability that at least one component fails is

$$P(X \geq 1) = p_X(1) + p_X(2) = 0.18 + 0.01 = 0.19.$$

Equivalently, the complement rule gives $P(X \geq 1) = 1 - p_X(0) = 1 - 0.81 = 0.19$.

The notation $p_X(x)$ identifies both the random variable and the value being considered. When several random variables occur in the same analysis, the subscript prevents ambiguity. The symbol p without a subscript is often used for a model parameter, so keeping the subscript on a PMF is useful.

Example 3.3 Determining an Unknown PMF Constant

Suppose a discrete random variable X has support $\{0, 1, 2, 3\}$ and PMF

$$p_X(x) = c(x + 1), \qquad x \in \{0, 1, 2, 3\},$$

where c is an unknown constant. Since the probability masses must sum to one,

$$\sum_{x=0}^{3} p_X(x) = c(1+2+3+4) = 10c = 1.$$

Therefore, $c = 0.1$, and the PMF values are

x	0	1	2	3
$p_X(x)$	0.1	0.2	0.3	0.4.

For example,

$$P(X \geq 2) = p_X(2) + p_X(3) = 0.3 + 0.4 = 0.7.$$

The cumulative distribution function

A PMF assigns probability to individual values. The cumulative distribution function instead accumulates all probability at or below a threshold. Unlike a PMF, a CDF is defined for every real number and can therefore be used for discrete as well as continuous random variables.

Definition 3.4 (Cumulative Distribution Function)

The **cumulative distribution function** (CDF) of a random variable X is

$$F_X(x) = P(X \leq x), \qquad x \in \mathbb{R}.$$

For a discrete random variable, the CDF is obtained by summing the probability masses up to x:

$$F_X(x) = \sum_{t \leq x} p_X(t).$$

The summation includes only values t in the support of X. Consequently, the CDF of a discrete random variable is a step function. It remains constant between successive support values and jumps by $p_X(x)$ at each value x in the support.

Theorem 3.2 (Properties of a Cumulative Distribution Function)

Every CDF F_X has the following properties:

1. $0 \leq F_X(x) \leq 1$ for every $x \in \mathbb{R}$;
2. F_X is non-decreasing: if $a < b$, then $F_X(a) \leq F_X(b)$;
3. F_X is right-continuous;
4. $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$.

These properties follow from the interpretation of $F_X(x)$ as accumulated probability. Raising the threshold cannot remove outcomes, and a sufficiently small or sufficiently large threshold eventually includes none or all of the possible values.

Example 3.4 CDF for Failed Components

For the number of failed components, the PMF values are 0.81, 0.18, and 0.01 at 0, 1, and 2, respectively.

Therefore,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.81, & 0 \le x < 1, \\ 0.99, & 1 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

At $x = 1$, for example,

$$F_X(1) = P(X \le 1) = p_X(0) + p_X(1) = 0.81 + 0.18 = 0.99.$$

The CDF jumps by 0.81 at 0, by 0.18 at 1, and by 0.01 at 2. Each jump is exactly the corresponding PMF value.

The CDF is especially convenient for threshold and interval probabilities. For any real numbers $a < b$,

$$P(a < X \le b) = F_X(b) - F_X(a).$$

Tail probabilities follow from the complement rule:

$$P(X > x) = 1 - F_X(x).$$

Care is required with strict and non-strict inequalities for a discrete random variable because an individual value can have positive probability. For an integer-valued random variable,

$$P(X \ge k) = 1 - F_X(k-1),$$

not $1 - F_X(k)$.

Example 3.5 Probabilities from a CDF

Return to the random variable with PMF

$$p_X(0) = 0.1, \qquad p_X(1) = 0.2, \qquad p_X(2) = 0.3, \qquad p_X(3) = 0.4.$$

Its CDF at the support values is

$$F_X(0) = 0.1, \qquad F_X(1) = 0.3, \qquad F_X(2) = 0.6, \qquad F_X(3) = 1.$$

The probability that X is strictly greater than 1 is

$$P(X > 1) = 1 - F_X(1) = 1 - 0.3 = 0.7.$$

The probability that X lies between 1 and 3, excluding 1 but including 3, is

$$P(1 < X \le 3) = F_X(3) - F_X(1) = 1 - 0.3 = 0.7.$$

If the lower endpoint is included instead, then

$$P(1 \le X \le 3) = 1 - F_X(0) = 1 - 0.1 = 0.9.$$

The difference occurs because $P(X = 1) = 0.2$.

The PMF and CDF contain the same information for a discrete random variable. The CDF is obtained by accumulating the PMF, and the PMF is recovered from the jumps of the CDF. Writing

$$F_X(x^-) = \lim_{t \uparrow x} F_X(t)$$

for the value immediately to the left of x, we have

$$p_X(x) = F_X(x) - F_X(x^-).$$

For an integer-valued random variable, this becomes

$$p_X(k) = F_X(k) - F_X(k-1).$$

The choice between PMF and CDF therefore depends on the question. The PMF is natural for exact values such as $P(X = 2)$, whereas the CDF is often more convenient for thresholds, ranges, and tail probabilities.

3.2 Expectation and variance

A probability distribution describes all possible values of a random variable and the probability attached to each value. For many purposes, however, it is useful to summarise the distribution using a small number of quantities. The **expected value** describes its centre, while the **variance** and **standard deviation** describe how widely its probability is spread around that centre.

These quantities are properties of a probability model. They should therefore be distinguished from the descriptive measures introduced in chapter 1. The expected value $E[X]$ is the theoretical mean of the random variable X, whereas a sample mean $\bar{x}$ is calculated from observed data. Similarly, $\mathrm{Var}(X)$ describes variation in the probability distribution, whereas a sample variance s^2 is calculated from a particular sample.

Expected value

Suppose that a discrete random variable X has support $\{x_1, x_2, \ldots\}$. Its expected value is a probability-weighted average of these possible values. Values with high probability contribute more than values with low probability.

Definition 3.5 (Expected Value of a Discrete Random Variable)

Let X be a discrete random variable with PMF p_X. The **expected value**, or **mean**, of X is

$$E[X] = \sum_{x \in \mathrm{supp}(X)} x\, p_X(x),$$

provided that the sum exists. The expected value is also denoted by μ_X or simply μ when the random variable is clear from context. ♣

The expected value is not necessarily a possible value of the random variable. It describes the long-run average obtained by repeating the random experiment under the same conditions, not the result expected in any one trial.

Example 3.6 Expected Number of Failed Components

For the number X of failed components introduced in section 3.1, the PMF is

x	0	1	2
$p_X(x)$	0.81	0.18	0.01.

The expected number of failed components is therefore

$$\begin{aligned} E[X] &= 0 \cdot 0.81 + 1 \cdot 0.18 + 2 \cdot 0.01 \\ &= 0.20. \end{aligned}$$

This does not mean that 0.20 of a component will fail in a particular inspection. It means that over many inspections of two components, the average number of failures per inspection will approach 0.20 if the model is appropriate. For example, across 1000 such inspections, the model predicts approximately $1000 \cdot 0.20 = 200$ failed components in total.

Example 3.7 Expected Value of a Fair Die

Let X be the number shown on a fair six-sided die. Since every value has probability $1/6$,

$$\begin{aligned} E[X] &= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \cdots + 6 \cdot \frac{1}{6} \\ &= \frac{1+2+3+4+5+6}{6} \\ &= 3.5. \end{aligned}$$

No single roll can produce 3.5. The value is nevertheless the centre of the probability distribution and the long-run average of repeated rolls.

Expected value is linear. Multiplying every possible value by a constant multiplies the expected value by the same constant, and adding a constant shifts the expected value by that amount.

Theorem 3.3 (Linearity of Expected Value)

If X is a random variable with a finite expected value and a and b are constants, then

$$E[aX + b] = aE[X] + b.$$

♡

More generally, an engineering quantity of interest may be a function of a random variable. If $Y = g(X)$, it is not necessary to determine the complete PMF of Y before calculating its expectation. Each possible value $g(x)$ can be weighted directly by the probability $p_X(x)$.

Theorem 3.4 (Expectation of a Function of a Random Variable)

Let X be a discrete random variable with PMF p_X, and let g be a real-valued function. Then

$$E[g(X)] = \sum_{x \in \text{supp}(X)} g(x) p_X(x),$$

provided that the sum exists.

♡

The linearity result follows by choosing $g(x) = ax + b$. The more general form is useful when costs, energy consumption, response penalties, or other consequences depend nonlinearly on the outcome.

Example 3.8 Expected Inspection Cost

Suppose that inspecting a pair of components has a fixed setup cost of EUR 50, and every failed component produces an additional cost of EUR 600. If X is the number of failed components, the total cost is

$$C = 50 + 600X.$$

Using linearity and the result $E[X] = 0.20$,

$$E[C] = 50 + 600E[X] = 50 + 600 \cdot 0.20 = 170.$$

Thus, the expected cost is EUR 170 per inspection. This is a long-run average cost; an individual inspection costs EUR 50, EUR 650, or EUR 1250 depending on whether zero, one, or two components fail.

Example 3.9 Expected Loss under a Nonlinear Penalty

Suppose instead that a simultaneous failure of both components is especially disruptive. Let $L = g(X)$ be the loss, where $g(0) = 0$, $g(1) = 500$, and $g(2) = 5000$, measured in euros. The expected loss is

$$\begin{aligned} E[L] &= g(0)p_X(0) + g(1)p_X(1) + g(2)p_X(2) \\ &= 0 \cdot 0.81 + 500 \cdot 0.18 + 5000 \cdot 0.01 \\ &= 140. \end{aligned}$$

Using $g(E[X])$ would not give the correct result: the penalty is nonlinear, and $g(0.20)$ is not even defined by the stated loss table. The probabilities must be applied to the possible losses themselves.

Variance and standard deviation

Two distributions can have the same expected value while assigning probability very differently around it. Expected value alone therefore does not describe the uncertainty in a random variable. Variance measures the expected squared distance from the mean.

Definition 3.6 (Variance and Standard Deviation)

Let X be a random variable with mean $\mu_X = E[X]$. The **variance** of X is

$$\operatorname{Var}(X) = E\big[(X - \mu_X)^2\big].$$

For a discrete random variable,

$$\operatorname{Var}(X) = \sum_{x \in \operatorname{supp}(X)} (x - \mu_X)^2 p_X(x).$$

The **standard deviation** of X is

$$\sigma_X = \sqrt{\operatorname{Var}(X)}.$$

♣

The distances from the mean are squared for two reasons. First, positive and negative deviations cannot cancel. Second, large deviations receive greater weight than small deviations. Variance is always non-negative and equals zero only when X is constant with probability one.

Because the deviations are squared, variance has squared units. If X is measured in milliseconds, then $\operatorname{Var}(X)$ is measured in square milliseconds. Standard deviation has the same unit as X and is therefore usually easier to interpret directly.

The definition of variance is conceptually important, but the following equivalent formula is often more convenient for calculation.

Theorem 3.5 (Computational Formula for Variance)

If $E[X^2]$ is finite, then

$$\operatorname{Var}(X) = E[X^2] - \big(E[X]\big)^2.$$

♡

Example 3.10 Variance of the Number of Failed Components

For the component model, $E[X] = 0.20$. To use the computational formula, first calculate

$$\begin{aligned} E[X^2] &= 0^2 \cdot 0.81 + 1^2 \cdot 0.18 + 2^2 \cdot 0.01 \\ &= 0.22. \end{aligned}$$

Therefore,

$$\operatorname{Var}(X) = E[X^2] - \big(E[X]\big)^2 = 0.22 - 0.20^2 = 0.18,$$

and

$$\sigma_X = \sqrt{0.18} \approx 0.424.$$

The expected number of failures is 0.20 per inspection, and the standard deviation is approximately 0.424 failures.

Variance responds differently from expectation to a linear transformation. Adding a constant shifts every possible value and the mean by the same amount, so it does not change the distances from the mean. Multiplication by a multiplies every distance by a and every squared distance by a^2.

Theorem 3.6 (Variance under a Linear Transformation)

If X has finite variance and a and b are constants, then

$$\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

Consequently,

$$\sigma_{aX+b} = |a|\sigma_X.$$

♡

Example 3.11 Variation in Inspection Cost

For the inspection cost $C = 50 + 600X$, the fixed setup cost changes the mean but contributes no variation. Hence,

$$\begin{aligned} \operatorname{Var}(C) &= 600^2 \operatorname{Var}(X) \\ &= 600^2 \cdot 0.18 \\ &= 64\,800, \end{aligned}$$

measured in square euros. The standard deviation is

$$\sigma_C = 600\sqrt{0.18} \approx 254.56 \text{ EUR}.$$

Thus, the model gives an expected cost of EUR 170 with a standard deviation of approximately EUR 255. The standard deviation is large relative to the expected cost because the usually low cost is accompanied by a small probability of a much larger failure cost.

The mean and variance do not determine the distribution

Expected value and variance provide useful summaries, but they do not replace the PMF or CDF. Even distributions with the same mean and variance can differ in other ways, and distributions with the same mean can have very different levels of uncertainty.

Example 3.12 Same Mean, Different Variability

Consider two random variables X and Y with PMFs

x	0	1	2
$p_X(x)$	0.25	0.50	0.25
$p_Y(x)$	0.50	0	0.50.

Both variables have expected value one:

$$E[X] = 0 \cdot 0.25 + 1 \cdot 0.50 + 2 \cdot 0.25 = 1$$

and

$$E[Y] = 0 \cdot 0.50 + 1 \cdot 0 + 2 \cdot 0.50 = 1.$$

For X,

$$E[X^2] = 0^2 \cdot 0.25 + 1^2 \cdot 0.50 + 2^2 \cdot 0.25 = 1.5,$$

so

$$\mathrm{Var}(X) = 1.5 - 1^2 = 0.5.$$

For Y,

$$E[Y^2] = 0^2 \cdot 0.50 + 1^2 \cdot 0 + 2^2 \cdot 0.50 = 2,$$

so

$$\mathrm{Var}(Y) = 2 - 1^2 = 1.$$

Although the two models have the same expected value, Y never takes the central value 1 and places all its probability at the two extremes. Its variance is therefore twice as large. A decision based only on the expected value would overlook this difference in uncertainty.

Expectation and variance will be calculated for each of the named distributions in the following sections. Their values help describe and compare the models, but the choice of model must still be based on the underlying random experiment and the complete form of the distribution.

3.3 Bernoulli and binomial models

Many engineering measurements begin with a binary question. Did a request complete successfully? Was an intrusion attempt blocked? Did the temperature exceed a specified threshold? Was a delivery late? Did a bearing fail, a casting pass inspection, or a structural element satisfy its load requirement? Each question has two possible answers. A **Bernoulli model** describes one such trial, while a **binomial model** describes the number of successes in a fixed collection of repeated Bernoulli trials.

The words *success* and *failure* are conventional labels only. A success is the outcome coded as 1 and may represent an undesirable event such as a defect, packet loss, late delivery, or security alert. What matters is that the event of interest is defined clearly before the probability model is constructed.

Bernoulli trials

A single trial with two possible outcomes can be represented numerically by assigning the value 1 to the event of interest and 0 to its complement. This numerical representation is also called an indicator random variable.

Definition 3.7 (Bernoulli Random Variable)

A random variable X has a **Bernoulli distribution** with parameter p if

$$P(X = 1) = p \qquad \text{and} \qquad P(X = 0) = 1 - p,$$

where $0 \le p \le 1$. We write

$$X \sim \text{Bernoulli}(p).$$

The value 1 denotes success and the value 0 denotes failure. ♣

The PMF can be written compactly as

$$p_X(x) = p^x (1-p)^{1-x}, \qquad x \in \{0, 1\}.$$

When $x = 1$, this expression equals p; when $x = 0$, it equals $1 - p$. The CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

Theorem 3.7 (Mean and Variance of a Bernoulli Random Variable)

If $X \sim \text{Bernoulli}(p)$, then

$$E[X] = p, \qquad \text{Var}(X) = p(1-p).$$

♡

The variance is zero when $p = 0$ or $p = 1$ because the outcome is then certain. It is largest when $p = 0.5$, where the two outcomes are equally likely and the uncertainty about a single trial is greatest.

Example 3.13 Cybersecurity Alert for One Authentication Attempt

Suppose a monitoring rule flags an authentication attempt with probability 0.03 under a specified operating condition. Let

$$X = \begin{cases} 1, & \text{the attempt is flagged,} \\ 0, & \text{the attempt is not flagged.} \end{cases}$$

Then $X \sim \text{Bernoulli}(0.03)$, and

$$P(X = 1) = 0.03, \qquad P(X = 0) = 0.97.$$

Its mean and variance are

$$E[X] = 0.03$$

and

$$\text{Var}(X) = 0.03(1 - 0.03) = 0.0291.$$

The expected value 0.03 is also the long-run proportion of attempts that the model predicts will be flagged. It is not, by itself, the probability that a flagged attempt is genuinely malicious; that would require information about the accuracy of the rule and the prevalence of malicious attempts, as discussed in section 2.7.

From Bernoulli trials to a binomial count

Suppose the same type of Bernoulli trial is repeated n times and that X counts the number of successes. A binomial model is appropriate only when the structure of the repeated experiment satisfies specific assumptions.

Definition 3.8 (Binomial Experiment)

A **binomial experiment** consists of trials satisfying all four conditions:

1. the number of trials n is fixed in advance;
2. each trial has exactly two outcomes, labelled success and failure;
3. the probability of success is the same value p on every trial; and
4. the trials are independent.

The random variable of interest is the number of successes among the n trials. ♣

The assumptions describe the data-generating process, not merely the observed data. Seeing 7 failures in 100 tests does not establish a binomial model. We must also have a reason to regard the 100 tests as a fixed number of independent trials with a common failure probability.

The binomial coefficient

To derive the probability of exactly k successes, first consider one particular ordering. For example, the sequence

$$SSFSS$$

contains four successes and one failure. If the trials are independent, the probability of this particular sequence is

$$p^4(1 - p).$$

Every other sequence with four successes and one failure has the same probability. We therefore need to count how many different sequences contain exactly k successes among n positions.

Definition 3.9 (Binomial Coefficient)

For integers $0 \le k \le n$, the **binomial coefficient**

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

is the number of ways to choose the k positions occupied by successes among n positions. By convention, $0! = 1$.

♣

Example 3.14 Placing Two Defects among Five Units

Suppose five units are inspected and exactly two are defective. Once the two positions containing defects have been chosen, the entire defect pattern is determined. The number of possible patterns is

$$\binom{5}{2} = \frac{5!}{2!3!} = 10.$$

If each unit is independently defective with probability p, every one of these ten patterns has probability

$$p^2(1-p)^3.$$

Consequently, the total probability of exactly two defective units is

$$10p^2(1-p)^3.$$

This is the only counting principle needed for the binomial model: $\binom{n}{k}$ counts the possible placements of k successes, while independence gives the probability of each placement.

The binomial distribution

Combining the number of possible placements with the probability of each placement gives the binomial PMF.

Definition 3.10 (Binomial Distribution)

Let X be the number of successes in n independent Bernoulli trials, each with success probability p. Then X has a **binomial distribution**, written

$$X \sim \mathrm{Bin}(n, p),$$

with PMF

$$p_X(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

♣

The four factors in the PMF have distinct meanings:

$$\underbrace{\binom{n}{k}}_{\text{placements}} \underbrace{p^k}_{\text{successes}} \underbrace{(1-p)^{n-k}}_{\text{failures}}.$$

The support is $\{0, 1, \ldots, n\}$. For an integer k in this support, the CDF is

$$F_X(k) = P(X \le k) = \sum_{j=0}^{k} \binom{n}{j} p^j (1-p)^{n-j}.$$

Theorem 3.8 (Mean and Variance of a Binomial Random Variable)

If $X \sim \text{Bin}(n, p)$, then

$$E[X] = np, \qquad \text{Var}(X) = np(1-p),$$

and

$$\sigma_X = \sqrt{np(1-p)}.$$

♡

The mean has a direct interpretation. Each of the n trials contributes a success with probability p, so the expected number of successes is n times p. The variance also adds across the independent Bernoulli trials, giving n times the Bernoulli variance $p(1-p)$. Independence is essential for this variance result.

Example 3.15 Defective Units in a Production Batch

Suppose a production process has a defect probability of 0.03 for each unit. Twenty units are selected under conditions that make their defect indicators reasonably independent, and X denotes the number of defective units. Under the stated assumptions,

$$X \sim \text{Bin}(20, 0.03).$$

The probability of exactly one defective unit is

$$\begin{aligned} P(X = 1) &= \binom{20}{1}(0.03)^1(0.97)^{19} \\ &\approx 0.3364. \end{aligned}$$

The expected number and standard deviation are

$$E[X] = 20 \cdot 0.03 = 0.60$$

and

$$\sigma_X = \sqrt{20 \cdot 0.03 \cdot 0.97} \approx 0.763.$$

Thus, the model predicts an average of 0.60 defective units per group of twenty. This does not imply that most inspected groups contain exactly 0.60 defects, which is not a possible count.

Exact, cumulative, and tail probabilities

The wording of a problem determines whether the PMF, CDF, or complement rule is most convenient. For $X \sim \text{Bin}(n, p)$,

$$P(X = k) = p_X(k)$$

is an exact probability,

$$P(X \le k) = F_X(k)$$

is cumulative, and

$$P(X \ge k) = 1 - F_X(k-1)$$

is an upper-tail probability. For integers $a \le b$,

$$P(a \le X \le b) = F_X(b) - F_X(a-1).$$

Careful attention to the endpoints prevents the off-by-one errors discussed for general CDFs in section 3.1.

Example 3.16 Packet Loss in a Software System

Suppose a message contains 100 packets. Under a simplified model, each packet is lost independently with probability 0.01, and X is the number of packets that must be retransmitted. Then

$$X \sim \text{Bin}(100, 0.01).$$

The probability that at least one packet must be retransmitted is most easily found using the complement:

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - \binom{100}{0}(0.01)^0(0.99)^{100} \\ &= 1 - (0.99)^{100} \\ &\approx 0.6340. \end{aligned}$$

For at least two retransmissions, subtract the probabilities of zero and one:

$$\begin{aligned} P(X \geq 2) &= 1 - P(X = 0) - P(X = 1) \\ &= 1 - (0.99)^{100} - \binom{100}{1}(0.01)(0.99)^{99} \\ &\approx 0.2642. \end{aligned}$$

The mean and standard deviation are

$$E[X] = 100 \cdot 0.01 = 1$$

and

$$\sigma_X = \sqrt{100 \cdot 0.01 \cdot 0.99} \approx 0.995.$$

Although only one packet is lost on average, the probability of at least one retransmission is approximately 63.4%. Expected counts and event probabilities answer different questions.

Example 3.17 Late Deliveries in a Supply Chain

A supplier schedules 12 deliveries during a planning period. Suppose each delivery is late with probability 0.08, independently of the others, and let X be the number of late deliveries. Then

$$X \sim \text{Bin}(12, 0.08).$$

The probability of at most one late delivery is

$$\begin{aligned} P(X \leq 1) &= P(X = 0) + P(X = 1) \\ &= (0.92)^{12} + \binom{12}{1}(0.08)(0.92)^{11} \\ &\approx 0.7513. \end{aligned}$$

The probability of exactly two late deliveries is

$$P(X = 2) = \binom{12}{2}(0.08)^2(0.92)^{10}$$
$$\approx 0.1835.$$

The expected number of late deliveries is

$$E[X] = 12 \cdot 0.08 = 0.96.$$

These calculations are meaningful only if a common delay mechanism, such as the same port closure or severe weather system, does not create substantial dependence among deliveries.

Example 3.18 Structural Fastener Tests

Twelve structural fasteners are subjected to identical acceptance tests. Suppose each fastener passes with probability 0.98, independently of the others, and let X be the number that pass. Then

$$X \sim \mathrm{Bin}(12, 0.98).$$

The probability that all fasteners pass is

$$P(X = 12) = (0.98)^{12} \approx 0.7847.$$

The probability that at least eleven pass is

$$P(X \geq 11) = P(X = 11) + P(X = 12)$$
$$= \binom{12}{11}(0.98)^{11}(0.02) + (0.98)^{12}$$
$$\approx 0.9769.$$

Requiring every fastener to pass is therefore much more stringent than allowing one failure: the corresponding probabilities are approximately 78.5% and 97.7% under this model.

Assessing the binomial assumptions

A binomial calculation can be numerically correct and still be an inappropriate analysis if its assumptions do not match the engineering process. Each assumption should be connected to a feature of the application.

Example 3.19 Can Exceedance Days Be Modelled as Binomial?

An environmental engineer records whether a river exceeds a warning level on each of 30 consecutive days and proposes the model

$$X \sim \text{Bin}(30, p),$$

where X is the number of exceedance days. The number of trials is fixed and every day has two coded outcomes. However, the remaining assumptions require scrutiny.

Exceedances on consecutive days are likely to be dependent because river levels persist and several days may be influenced by the same storm. The probability of exceedance may also vary with season, upstream releases, and soil saturation. Consequently, a binomial model is not justified merely because the response is binary on each day. It may be a rough approximation under stable conditions and weak dependence, but those conditions must be argued rather than assumed.

Dependence also arises through common-cause mechanisms. Fasteners installed in the same concrete pour may share material and installation conditions. Software requests handled by the same failing service instance may fail together. Deliveries routed through the same port may be delayed together. In each case, treating the outcomes as independent usually understates the chance of clusters of failures.

The constant-probability assumption can fail when equipment wears, operators learn, weather changes, or suppliers differ. Sampling without replacement from a small finite lot creates another problem: after a defective item is selected, the composition of the remaining lot changes. For example, drawing ten bearings without replacement from a lot of fifty containing four defective bearings does not produce independent Bernoulli trials with a constant defect probability. A binomial model may sometimes be an adequate approximation when the sampled fraction is small, but it is not the exact model.

The Bernoulli and binomial distributions therefore provide more than formulas. They express a particular structure: one binary trial, or a fixed number of comparable and independent binary trials. In the next section, the focus shifts from a fixed number of trials to events counted over an interval of time, space, area, volume, or another exposure measure.

3.4 Poisson models

The binomial distribution counts successes among a fixed number of trials. Many engineering counts arise differently. A server may experience an uncertain number of failures during a day, a sheet of material may contain an uncertain number of surface flaws, or a warehouse may receive an uncertain number of urgent orders during an hour. In these situations, the number of opportunities is not naturally fixed in advance. Instead, events are counted within a specified **exposure**, such as time, distance, area, volume, number of transactions, or operating hours.

The **Poisson distribution** is a fundamental model for such counts. It is governed by one parameter, λ, which is the expected number of events in the stated exposure. The exposure is part of the parameter definition: an expected count of two events per hour is not the same model as an expected count of two events per day.

When a Poisson model is plausible

A Poisson model is often considered when the following features are reasonable approximations:

1. X counts identifiable events in a clearly specified exposure;
2. the average event rate is approximately stable throughout that exposure;
3. occurrences do not substantially increase or decrease the chance of further occurrences, so clustering and inhibition are limited; and
4. there is no small fixed upper bound on the count that is central to the problem.

These are modelling conditions, not facts implied by observing count data. A column of non-negative integers is not automatically Poisson distributed. The mechanism producing the counts and the definition of the exposure must support the model.

Definition 3.11 (Poisson Distribution)

A discrete random variable X has a **Poisson distribution** with parameter $\lambda > 0$ if

$$P(X = k) = p_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

We write

$$X \sim \text{Pois}(\lambda).$$

♣

Here, $e \approx 2.71828$ is the base of the natural logarithm, and $0! = 1$. The support is the set of all non-negative integers. Although the mathematical model has no upper bound, very large counts may have extremely small probabilities when λ is modest.

The parameter λ is both the centre of the distribution and an expected count for one specific exposure. If a system has a stable average rate r events per unit of exposure and the exposure has size t, then

$$\lambda = rt.$$

The units cancel in the multiplication, but they remain essential to the interpretation. For example,

$$\left(0.8\ \frac{\text{failures}}{\text{hour}}\right)(4\ \text{hours}) = 3.2\ \text{failures}.$$

Thus, the appropriate Poisson parameter for four hours is 3.2, not 0.8.

PMF, CDF, and tail probabilities

The PMF gives the probability of an exact count. For $X \sim \text{Pois}(\lambda)$,

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}.$$

The CDF is obtained by summing the PMF:

$$F_X(k) = P(X \le k) = \sum_{j=0}^{k} \frac{e^{-\lambda}\lambda^j}{j!}.$$

As with the binomial distribution, complements are often the most efficient way to calculate upper-tail probabilities:

$$P(X \ge k) = 1 - F_X(k-1).$$

In particular,

$$P(X = 0) = e^{-\lambda}$$

and

$$P(X \ge 1) = 1 - e^{-\lambda}.$$

Example 3.20 Software Failures during an Operating Window

A service experiences failures at an average rate of 0.8 per hour under stable operating conditions. Let X be the number of failures during a four-hour monitoring window. The parameter for this exposure is

$$\lambda = \left(0.8 \frac{\text{failures}}{\text{hour}}\right)(4 \text{ hours}) = 3.2 \text{ failures},$$

so

$$X \sim \text{Pois}(3.2).$$

The probability of no failures is

$$P(X = 0) = e^{-3.2} \approx 0.0408.$$

The probability of at most two failures is

$$\begin{aligned} P(X \le 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= e^{-3.2}\left(1 + 3.2 + \frac{3.2^2}{2!}\right) \\ &\approx 0.3799. \end{aligned}$$

Therefore,

$$P(X \ge 3) = 1 - P(X \le 2) \approx 0.6201.$$

Under this model, three or more failures during the four-hour window are more likely than fewer than three. Whether that rate is operationally acceptable is an engineering decision, not a conclusion supplied by the probability model alone.

Mean and variance

The Poisson distribution has the distinctive property that its mean and variance are equal.

Theorem 3.9 (Mean and Variance of a Poisson Random Variable)

If $X \sim \text{Pois}(\lambda)$, then

$$E[X] = \lambda, \qquad \text{Var}(X) = \lambda,$$

and

$$\sigma_X = \sqrt{\lambda}.$$

♡

The equality of mean and variance is a property of the theoretical model. It does not mean that the sample mean and sample variance calculated from a finite dataset must be exactly equal. Their relationship can nevertheless be useful when assessing whether a Poisson model is plausible.

Example 3.21 Surface Flaws in Manufactured Panels

Suppose surface flaws occur at an average rate of 0.4 per square metre, and a panel has an area of 5 square metres. If the rate is stable across the surface and flaws occur without substantial clustering, then

$$\lambda = \left(0.4 \, \frac{\text{flaws}}{\text{m}^2}\right)(5 \text{ m}^2) = 2 \text{ flaws}.$$

Let X be the number of flaws on one panel. Then

$$X \sim \text{Pois}(2).$$

The probability of exactly three flaws is

$$\begin{aligned} P(X = 3) &= \frac{e^{-2}2^3}{3!} \\ &\approx 0.1804. \end{aligned}$$

The probability of at least one flaw is

$$P(X \geq 1) = 1 - e^{-2} \approx 0.8647.$$

Finally,

$$E[X] = 2, \qquad \sigma_X = \sqrt{2} \approx 1.414.$$

The model therefore predicts two flaws per panel on average, with a standard deviation of approximately 1.41 flaws.

Changing the exposure

When the rate is constant, changing the exposure changes λ proportionally. If the event rate is r per unit and the exposure changes from t_1 to t_2, then

$$\lambda_1 = rt_1 \qquad \text{and} \qquad \lambda_2 = rt_2.$$

It is incorrect to reuse a parameter defined for one exposure without rescaling it to the exposure in the probability question.

Example 3.22 Urgent Orders during Twenty Minutes

A distribution centre receives urgent orders at an average rate of six per hour. Let X be the number received during the next twenty minutes. Since twenty minutes is one third of an hour,

$$\lambda = \left(6 \ \frac{\text{orders}}{\text{hour}}\right)\left(\frac{1}{3}\ \text{hour}\right) = 2 \text{ orders}.$$

Thus, $X \sim \text{Pois}(2)$. The probability of more than four urgent orders is

$$\begin{aligned} P(X > 4) &= 1 - P(X \leq 4) \\ &= 1 - \sum_{k=0}^{4} \frac{e^{-2}2^k}{k!} \\ &\approx 0.0527. \end{aligned}$$

The probability of more than four arrivals in twenty minutes is therefore approximately 5.3%. Using $\lambda = 6$ would answer a different question: the number of arrivals in one hour.

Poisson approximation to the binomial distribution

The Poisson distribution can also approximate a binomial distribution when the number of trials is large, the success probability is small, and

$$\lambda = np$$

is of moderate size. The approximation replaces

$$X \sim \text{Bin}(n, p)$$

with

$$X \approx \text{Pois}(np).$$

It is especially useful for rare-event counts when evaluating the binomial PMF directly is inconvenient. The approximation does not remove the binomial assumptions: the underlying trials must still be independent with a common success probability.

Example 3.23 Rare Security Incidents

Suppose $10\,000$ independent connection attempts each have probability 0.0002 of producing a particular security incident. If X is the number of incidents, the exact model is

$$X \sim \text{Bin}(10\,000, 0.0002).$$

Since n is large and p is small, use the Poisson parameter

$$\lambda = np = 10\,000 \cdot 0.0002 = 2.$$

The probability of at least three incidents is approximately

$$\begin{aligned} P(X \geq 3) &\approx 1 - P(Y \leq 2), \qquad Y \sim \text{Pois}(2), \\ &= 1 - e^{-2}\left(1 + 2 + \frac{2^2}{2!}\right) \\ &\approx 0.3233. \end{aligned}$$

The exact binomial probability is also 0.3233 to four decimal places in this case. Such close agreement is not guaranteed when p is larger or when n is small.

Assessing the Poisson assumptions

The mean–variance relationship provides a useful diagnostic. If repeated counts from comparable exposures have a sample variance substantially larger than their sample mean, the data are said to show **overdispersion** relative to a Poisson model. Possible causes include clustering, unobserved differences between exposures, a changing event rate, or common-cause failures. A sample variance substantially smaller than the sample mean is called **underdispersion** and may arise from spacing, inhibition, capacity limits, or unusually regular processes.

Neither pattern proves that a Poisson model is wrong, because sample summaries vary from one dataset to another. They identify features that should be investigated using process knowledge, visualisation, and later statistical methods.

Example 3.24 Annual Flood-Warning Events

Suppose annual counts of flood-warning events over comparable years have a sample mean of 3.5 and a sample variance of 9.1. A Poisson model with $\lambda = 3.5$ would have theoretical variance 3.5, considerably smaller than the observed sample variance.

The difference suggests overdispersion. A plausible explanation is that some years are much wetter than others and that several warnings may arise from the same persistent weather system. The annual rate is then not stable across years, and events within a year may be clustered. The comparison is a diagnostic rather than a formal rejection of the model, but it gives a clear reason to examine the mechanism before using Poisson tail probabilities for planning.

A Poisson model is also inappropriate when the count has a small fixed maximum that matters. The number of failed components among twenty tested components cannot exceed twenty and naturally begins with a binomial model. By contrast, the number of failures during a period of operation may have no comparable fixed trial count and can be a plausible Poisson quantity. The model choice follows from the event-generating structure, not simply from whether the observed variable is a count.

Together, the Bernoulli, binomial, and Poisson distributions cover three common engineering structures: one binary outcome, successes among a fixed number of trials, and events within a specified exposure. Selecting among them requires both a correctly calculated probability and a credible account of how the data are generated.

The models in this chapter describe binary outcomes and event counts. Many engineering quantities, however, are measured on a continuous scale, including response times, dimensions, temperatures, and component lifetimes. The next chapter develops probability models for such continuous measurements.

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